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System and method for dynamic tow of a trailer

Abstract

One variation of a method includes, during a first time period: detecting a direction of motion of a trailer; detecting a first force applied to a kingpin; detecting an incline angle of the trailer; calculating a first target preload force opposite the direction of motion and inversely proportional to the incline angle; and in response to the first force falling below the first target preload force, triggering a motor to increase torque output opposite the direction of motion. The method further includes, during a second time period: detecting a second force applied to the kingpin; detecting a decline angle of the trailer; calculating a second target preload force opposite the direction of motion and inversely proportional to the decline angle; and in response to the second force falling below the second target preload force, triggering the motor to increase torque output opposite the direction of motion.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10189476	12/2018	Edren	N/A	B60L 15/2036
2023/0327475	12/2022	Ducher	701/36	B62D 53/0842

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application is a continuation application of U.S. patent application Ser. No. 18/238,408, filed on 25 Aug. 2023, which claims the benefit of U.S. Provisional Application No. 63/401,030, filed on 25 Aug. 2022, 63/420,469, filed on 28 Oct. 2022, and 63/431,273, filed on 8 Dec. 2022, each of which is incorporated in its entirety by this reference.

TECHNICAL FIELD

(1) This invention relates generally to the field of tow control and more specifically to a new and useful system and method for dynamic tow of a trailer in the field of tow control.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIGS. 1A, 1B, and 1C are schematic representations of a system;
- (2) FIGS. 2A, 2B, and 2C are schematic representations of one variation of the system;
- (3) FIGS. 3A, 3B, and 3C are schematic representations of one variation of the system;

- (4) FIG. 4 is a flowchart representation of a method;
- (5) FIGS. 5A, 5B, 5C, and 5D are schematic representations of one variation of the system; and
- (6) FIGS. 6A, 6B, 6C, 6D, and 6E are schematic representations of one variation of the system;
- (7) FIGS. 7A, 7B, and 7C are flowchart representations of one variation of the method;
- (8) FIGS. 8A, 8B, 8C, and 8D are schematic representations of one variation of the system;
- (9) FIG. 9 is a flowchart representation of one variation of the method; and
- (10) FIG. 10 is a schematic representation of one variation of the system.

DESCRIPTION OF THE EMBODIMENTS

(11) The following description of embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention. Variations, configurations, implementations, example implementations, and examples described herein are optional and are not exclusive to the variations, configurations, implementations, example implementations, and examples they describe. The invention described herein can include any and all permutations of these variations, configurations, implementations, example implementations, and examples.

(12) 1. System

(13) As shown in FIGS. 1A, 2A, 2B, 6B and 6C, a system **100** for dynamic tow of a trailer **140** includes: a kingpin **110**; and a controller **150**. The kingpin **110** includes: a head **117**; a base coupled to a proximal end of the trailer **140** and configured to transfer vertical loads from the trailer **140** into a hitch of a tow vehicle; a shank **116** interposed between the head **117** and the base and configured to transiently couple to the hitch of the tow vehicle; a first sensor **119** configured to output signals representing lateral forces applied to the kingpin **110**; and a second sensor **119** configured to output signals representing longitudinal forces applied to the kingpin **110**. The controller **150** is configured to: access a first signal from the first sensor **119** representing a first lateral force applied to the kingpin **110**; access a second signal from the second sensor **119** representing a first longitudinal force applied to the kingpin **110**; calculate a first direction and a first magnitude of a first force applied to the kingpin **110** based on the first signal and the second signal; and trigger a motor arranged on a distal end **149** of the trailer **140**, opposite the kingpin **110**, to output a first torque in the first direction of the first force and proportional to the first magnitude of the first force.

(14) 1.1 Variation: Integrated Trailer+Drive System

(15) One variation of the system **100**, shown in FIGS. 3A, 3B, and 3C, includes: a kingpin **110**; a trailer **140**; a drive system **130**; and a controller **150**. The kingpin **110** includes: a head **117**; a base coupled to a proximal end of the trailer **140** and configured to transfer vertical loads from the trailer **140** into a hitch of a tow vehicle; a shank **116** interposed between the head **117** and the base and configured to transiently couple to the hitch of the tow vehicle; a first sensor **119** configured to output signals representing lateral forces applied to the kingpin **110**; and a second sensor **119** configured to output signals representing longitudinal forces applied to the kingpin **110**.

(16) The trailer **140** includes: a floor **146**; a left rail **147** coupled to the floor **146**, extending parallel to and laterally offset from a longitudinal centerline **143** of the trailer **140**, and defining a first array of engagement features distributed along the left rail and longitudinally offset by a pitch distance; and a right rail **147** coupled to the floor **146**, extending parallel to and laterally offset from the longitudinal centerline **143** of the trailer **140** opposite the left rail **147**, and defining a second array of engagement features distributed along the right rail **147** and longitudinally offset by the pitch distance.

(17) The drive system **130** includes: a chassis; a set of latches configured to transiently engage a subset of engagement features, in the first array of engagement features on the left rail **147** and in the second array of engagement features in the right rail **147**, to retain the chassis on a distal end of the trailer **140** opposite the kingpin **110**; a passive axle **134** including a left passive wheel **135** and a right passive wheel **136**; a driven axle **137** adjacent the passive axle **134** and including a left driven

wheel **138** and a right driven wheel **139**; a motor **131** mounted to the chassis, coupled to the drive axle **137**; and configured to output torque to the left driven wheel **138** and to the right driven wheel **139**.

(18) 1.2 Variation: Kingpin+Controller

(19) One variation of the system **100** includes: a kingpin **110**; and a controller **150**. In this variation, the kingpin **110** includes: a head **117**; a base coupled to a proximal end of the trailer **140**; a shank **116** interposed between the head **117** and the base and configured to transiently couple to the hitch of the tow vehicle; and a set of sensors **119** configured to output a signal representing lateral forces and longitudinal forces applied to the kingpin **110**. The controller **150** is configured to: access a signal from the set of sensors **119** representing a first lateral force applied to the kingpin **110**; calculate a first direction and a first magnitude of a first force applied to the kingpin **110** by the hitch of the tow vehicle based on the signal; and trigger a motor **131** arranged on a distal end of the trailer **140**, opposite the kingpin **110**, to output a first torque in the first direction of the first force and proportional to the first magnitude of the first force.

(20) 1.3 Variation: Interface+Target Wheel Speed

(21) One variation of the system **100**, shown in FIG. 5B, includes: a kingpin **110**; an interface **120**; and a controller **150**. The kingpin **110** is: arranged on a proximal end **148** of the trailer **140**; includes a set of sensors **119** configured to output a signal representing lateral forces and longitudinal forces applied to the kingpin **110**; and configured to couple with a hitch of a tow vehicle. The interface **120** includes: a joystick **121**; and a kingpin **110** interface **123** configured to transiently couple to the kingpin **110** and to transfer forces, applied to the joystick **121**, into the kingpin **110**.

(22) In this variation of the system **100**, the controller **150** is configured to: access the signal from the set of sensors **119**; based on the signal, calculate a first direction and a first magnitude of a first force applied to the kingpin **110**; calculate a target left wheel speed of a left wheel **144** of the trailer **140** proportional to the first magnitude; calculate a target right wheel speed of a right wheel **145** of the trailer **140** proportional to the first magnitude, the target right wheel speed differing from the target left wheel speed based on the first direction; and serve the target left wheel speed and the target right wheel speed to a drive system **130** arranged proximal a distal end **149** of the trailer **140** opposite the kingpin **110**.

(23) 1.4 Variation: Manual Mode

(24) One variation of the system **100**, shown in FIG. 7A, includes: a kingpin **110**; an interface **120**; and a controller **150**. The kingpin **110**, arranged on a proximal end **148** of the trailer **140**; includes a first sensor **119** configured to output a first signal representing a first lateral force applied to the kingpin **110**; includes a second sensor **119** configured to output a second signal representing a first longitudinal force applied to the kingpin **110**; includes a third sensor **119** configured to output a third signal representing a first vertical force applied to the kingpin **110**; and is configured to couple with a hitch of a tow vehicle. The interface **120** includes a joystick **121** and a kingpin **110** interface **123** configured to transiently couple to the kingpin **110** and to transfer forces, applied to the joystick **121**, into the kingpin **110**.

(25) In this variation of the system **100**, the controller **150** is configured to, during an initial time period: access the third signal from the third sensor **119**; based on the third signal, detect the first vertical force as null; and, based on the first vertical force, activate a manual mode. The controller **150** is further configured to, during a first time period: access the first signal from the first sensor **119** and the second signal from the second sensor **119**; based on the first signal and the second signal, calculate a first direction and a first magnitude of a first force applied to the kingpin **110** via the joystick **121**; calculate a target left wheel speed of a left wheel of the trailer **140** proportional to the first magnitude; calculate a target right wheel speed of a right wheel of the trailer **140** proportional to the first magnitude; and serve the target left wheel speed and the target right wheel speed to a drive system **130** arranged on a distal end **149** of the trailer **140** opposite the kingpin

110.

(26) 1.5 Variation: Analog Joystick+Single Sensor

(27) One variation of the system **100** includes: a kingpin **110**; an interface **120**; and a controller **150**. The kingpin **110**: is arranged on a proximal end **148** of the trailer **140**; includes a sensor **119** configured to output a signal representing lateral and longitudinal forces applied to the kingpin **110**; and is configured to couple to a tow vehicle. The interface **120** includes an analog joystick; and a kingpin **110** interface **123** configured to transiently couple to the kingpin **110** and to transfer forces, applied to the analog joystick, into the kingpin **110**.

(28) In this variation of the system **100**, the controller **150** is configured to: access the signal from the sensor **119**; based on the signal, calculate a first magnitude of a first force applied to the kingpin **110**; calculate a first target speed of a first wheel of the trailer **140** proportional to the first magnitude; calculate a second target speed of a second wheel of the trailer **140** opposite the first wheel and proportional to the first magnitude; and serve the first target speed and the second target speed to a drive system **130** arranged on a distal end **149** of the trailer **140** opposite the kingpin **110**.

(29) 2. Applications

(30) Generally, the system **100** is configured to dynamically tow a trailer **140** (e.g., a towed vehicle) by: accessing signals from sensors **119** representing forces applied to a kingpin **110** arranged on a proximal end **148** of the trailer **140**; calculating a direction and a magnitude of each force applied to the kingpin **110** based on these signals; selectively entering a mode (e.g., a hibernation mode, a manual mode, a tow mode); identifying a coupling and/or a decoupling event between a hitch (e.g., a fifth wheel) of a tow vehicle (e.g., a tractor-trailer) and the kingpin **110** based on these forces; triggering a primary motor **131**, a set of secondary motors, a primary airbrake system **107**, and/or a set of secondary airbrake systems to manipulate driven wheels of the trailer **140** in a manual mode; calculating a target preload force as a function of a condition of the trailer **140** (e.g., a speed, an incline angle, a tractor-trailer **140** angle, a location, a charge state of a battery) in a tow mode; and triggering the primary motor **131**, arranged in a drive system located proximal a distal end of the trailer **140**, to selectively reduce torque output and/or increase torque output to decrease a difference between each force and the target preload force in the tow mode.

(31) 2.1 Hibernation Mode

(32) Furthermore, the system **100** can include a lockout system (e.g., a mechanical propulsion lockout system, an electrical lockout system) and the controller **150** can leverage the lockout system to prevent access to any sensor **119** data, voltage data, and/or controller **150** area network data (e.g., CAN data) by an unauthorized user, device, or tow vehicle and to prevent unauthorized access, modification, or theft of the kingpin **110** and/or the trailer **140**.

(33) Additionally, in a hibernation mode, the controller **150** can detect an acceleration of the trailer **140** at the kingpin **110**, identify an unauthorized movement event of the trailer **140**, and enter an antitheft mode. The controller **150** can then autonomously apply a regenerative brake force to the trailer **140** proportional to the acceleration. Thus, the controller **150** can enter an antitheft mode to prevent access to the kingpin **110** and/or the trailer **140** by an unauthorized user, device, or tow vehicle.

(34) The controller **150** can further implement a wireless communication protocol with a wireless transmitter arranged in the interface **120** and configured to broadcast an identifier to a wireless receiver arranged on the trailer **140** proximal the kingpin **110**. In response to verification of receipt of the identifier via the wireless receiver, the controller **150** can enter a tow mode to calculate a maximum wheel speed or enter a manual mode (e.g., tow vehicle less control of the trailer **140** or manually operated by a user or a remote control).

(35) 2.2 Manual Mode

(36) In manual mode, the system **100** includes an interface **120** including: a joystick **121** and a kingpin **110** interface **123** configured to transiently couple to the kingpin **110** and to transfer forces

(e.g., lateral forces, longitudinal forces, vertical forces), applied to the joystick **121** by a user (e.g., an operator, a yard manager) or a remote control **122**, into the kingpin **110**.

(37) Accordingly, the user (e.g., an operator, a yard manager) may manually manipulate the joystick **121** and the drive system **130** can then drive the wheels of the trailer **140** to relocate the trailer **140** according to forces applied to the joystick **121** by the user. Alternatively, the user may access a remote control **122** and set a target position for the trailer **140** and the drive system **130** can then drive the wheels of the trailer **140** to relocate the trailer **140** to the target position according to forces transferred to the joystick **121** by the remote control **122**.

(38) Furthermore, the drive system **130** can trigger a primary motor **131**, a set of secondary motors, a primary airbrake system **107**, and/or a set of secondary airbrake systems to manipulate the driven wheels of the trailer **140** and to relocate the trailer **140** to a new position. At the end of manual mode, the controller **150** can identify a coupling event at the kingpin **110** with the hitch of a tow vehicle, deactivate the manual mode, and enter a tow mode.

(39) 2.2 Tow Mode

(40) In tow mode, the controller **150** can detect conditions of the trailer **140** such as: a direction of motion of the trailer **140** (e.g., a forward direction, a reverse direction); a (e.g., a steering angle); a speed of the trailer **140**; an incline angle of the trailer **140** (e.g., a grade of a ground surface); a location of the trailer **140**; forces applied to the kingpin **110** (e.g., lateral forces, longitudinal forces, vertical forces, total forces); and a charge state of each battery pack coupled to the trailer **140** (e.g., a status, a level, a percentage). The controller **150** can then: calculate a target preload force proportional to and/or inversely proportional to the condition of the trailer **140**; and trigger a motor, arranged in the drive system, to increase torque output and/or reduce torque output in the direction of motion of the trailer **140** to decrease a difference between the target preload force and a total force applied to the kingpin **110** (e.g., an actual force) to control the trailer **140** in conjunction with the tow vehicle.

(41) Additionally, the controller **150** can monitor forces and accelerations of the trailer **140** at the kingpin **110** to detect abusive movement events in “real-time” and to set the target wheel speeds of the trailer **140** to null to slow motion of the trailer **140**, and thereby enable the user to maintain manual control of the joystick. At the end of tow mode, the controller **150** can identify a decoupling event at the kingpin **110** with the hitch of a tow vehicle, deactivate the tow mode, and then enter either the manual mode or the hibernation mode.

(42) Therefore, the controller **150** can leverage the dynamic target preload force, torque output, and regenerative braking: to prevent a jackknife event between the tow vehicle and the trailer **140** (e.g., tow vehicle brakes to a stop and the trailer **140** continues to move toward and push the tow vehicle or torque applied to the kingpin **110**); to reduce emissions by the tow vehicle; to decrease fuel consumption by the tow vehicle; to extend the life of each battery pack coupled to the trailer **140**; and to enable a driver of the tow vehicle to apply minimal to no brake force to the trailer **140** in tow mode.

(43) 3. Method

(44) As shown in FIGS. **4**, **7B**, and **9**, a method **S100** for dynamic tow of a trailer **140** includes, during a first time period: detecting a first longitudinal force applied to a kingpin **110**, arranged on a proximal end of a trailer **140**, by a hitch of a tow vehicle in Block **S110**; detecting a first lateral force applied to the kingpin **110** by the hitch in Block **S112**; detecting a first speed and a first direction of motion of the trailer **140** in Block **S120**; calculating a first tractor-trailer **140** angle, between longitudinal axes of the trailer **140** and the tow vehicle, based on the first longitudinal force and the first lateral force in Block **S130**; calculating a first total force, applied to the kingpin **110** by the hitch, based on the first longitudinal force and the first lateral force in Block **S132**; calculating a first target preload force opposite the first direction of motion and proportional to the first tractor-trailer **140** angle in Block **S140**; and triggering a motor, arranged in a drive system located proximal a distal end of the trailer **140**, to reduce torque output in the first direction of

motion to decrease a first difference between the first total force and the first target preload force in Block S150, in response to the first speed of the trailer 140 exceeding a first threshold speed and in response to the first total force falling below the first target preload force.

(45) The method S100 further includes, during a second time period: detecting a second longitudinal force applied to the kingpin 110 by the hitch in Block S110; detecting a second lateral force applied to the kingpin 110 by the hitch in Block S112; detecting a second speed and a second direction of the trailer 140 in Block S120; calculating a second total force applied to the kingpin 110 by the hitch based on the second longitudinal force and the second lateral force in Block S132; calculating a second target preload force opposite the second direction of motion and proportional to the second tractor-trailer 140 angle in Block S140; and triggering the motor, arranged in the drive system, to increase torque output in the second direction of motion to decrease a second difference between the second total force and the second target preload force in Block S160, in response to the second speed exceeding the first threshold speed and in response to the second total force exceeding the second target preload force.

(46) 3.1 Variation: Target Preload Force+Incline Angle

(47) One variation of the method S100 includes, during a first time period: detecting a direction of motion of the trailer 140 in Block S120; detecting a first force applied to a kingpin 110 arranged on a proximal end of the trailer 140, by a hitch of a tow vehicle in Block S132; detecting a first incline angle of the trailer 140 in Block S134; calculating a first target preload force opposite the direction of motion and inversely proportional to the first incline angle in Block S140; and, in response to the first force falling below the first target preload force, triggering a motor 131, arranged in a drive system 130 located proximal a distal end of the trailer 140, to increase torque output in the direction of motion to decrease a first difference between the first force and the first target preload force in Block S160.

(48) This variation of the method S100 further includes, during a second time period: detecting a second force applied to the kingpin 110 by the hitch of the tow vehicle in Block S132; detecting a decline angle of the trailer 140 in Block S134; calculating a second target preload force opposite the direction of motion of the tow vehicle and inversely proportional to the decline angle in Block S140; and, in response to the second force falling below the second target preload force, triggering the motor 131 to increase torque output opposite the direction of motion to decrease the second difference between the second force and the second target preload force in Block S160.

(49) 3.2 Variation: Target Preload Force+Tractor-Trailer Angle+Incline Angle

(50) One variation of the method S100 includes during a first time period: detecting a first force applied to a kingpin 110, arranged on a proximal end of a trailer 140, by a hitch of a tow vehicle in Block S132; detecting a first direction of motion of the trailer 140 in Block S120; detecting a first tractor-trailer 140 angle between longitudinal axes of the trailer 140 and the tow vehicle in Block S130; detecting a first incline angle of the trailer 140 in Block S134; calculating a first target preload force opposite the first direction of motion, proportional to the first tractor-trailer 140 angle, and inversely proportional to the first incline angle in Block S140; and, in response to the first force falling below the first target preload force, triggering a motor 131, arranged in a drive system 130 located proximal a distal end of the trailer 140, to reduce torque output in the first direction of motion to decrease a first difference between the first force and the first target preload force in Block S150.

(51) This variation of the method S100 further includes, during a second time period: detecting a second force applied to the kingpin 110 by the hitch in Block S132; detecting a second direction of motion of the trailer 140 in Block S120; detecting a second tractor-trailer 140 angle between longitudinal axes of the trailer 140 and the tow vehicle in Block S130; detecting a second incline angle of the trailer 140 in Block S134; calculating a second target preload force opposite the second direction of motion, proportional to the second tractor-trailer 140 angle, and inversely proportional to the second incline angle in Block S140; and, in response to the second force exceeding the

second target preload force, triggering the motor **131** to increase torque output in the second direction of motion to decrease a second difference between the second force and the second target preload force in Block **S160**.

(52) 4. Kingpin

(53) The kingpin **110** includes: a head **117**; a shank **116**; a base in; a set of fasteners **118**; a geolocation module; a wireless communications module; and a suite of sensors **119** including force sensors (e.g., a strain gauge, an IMU, a load cell), optical sensors (e.g., a one-dimensional depth sensor, a LIDAR sensor, an RGB camera), and/or inertial sensors (e.g., an IMU, an accelerometer, a gyroscope). The kingpin **110** is arranged on a proximal end **148** of a trailer **140**, is characterized by a unitary steel alloy structure, and is configured to interface with a hitch of a tow vehicle (e.g., a trailer **140** tractor, a semi, a semitruck), as shown in FIGS. **1A**, **1B**, **1C**, **2A**, **2B**, **2C**, **6A**, **6B**, and **6C**.

(54) In one implementation, the kingpin **110** is coupled to a floor **146** of a trailer **140** and is configured to transfer vertical loads from the trailer **140** into a hitch of a tow vehicle. In this implementation, the set of sensors **118** are configured to: output signals representing forces applied to the kingpin **110** (e.g., via the force sensors **119**); output signals representing inertial conditions of the trailer **140** (e.g., via the inertial sensors **119**); output signals representing a location of the trailer **140** (e.g., via the geolocation module); and transmit these force data, inertial conditions data, weight distribution data, and/or geolocation data to a controller **150** via the communications module.

(55) 4.1 Head+Shank+Base

(56) In one implementation, as shown in FIGS. **2A**, **2B**, and **2C**, the kingpin **110** includes: a head **117** defining a first diameter; a shank **116** defining a second diameter less than the first diameter; and the base in defining a third diameter greater than the first diameter of the head **117** and the second diameter of the shank **116**. In this implementation, the base in further defines a set of through-bores **114** arranged radially about the shank **116** and configured to receive a set of fasteners **118** to couple the kingpin **110** to a floor **146** of the trailer **140** and thus, fasten the kingpin **110** to the trailer **140**. Further, the shank **116** is configured to transiently couple to the hitch of the tow vehicle (e.g., a fifth wheel arranged on a tow vehicle).

(57) In one variation, the kingpin **110**: defines a first sensor receptacle extending parallel to a lateral axis of the trailer **140**; and defines a second sensor receptacle extending parallel to a longitudinal axis of the trailer **140**. Further, a first sensor (e.g., a strain gauge) is arranged in the first sensor receptacle and configured to output a signal representing shear forces in the kingpin **110** parallel to the lateral axis and a second sensor (e.g., a strain gauge) is arranged in the second sensor receptacle and configured to output the second signal representing shear forces in the kingpin **110** parallel to the longitudinal axis.

(58) In another variation, the shank **116** defines a first sensor receptacle extending parallel to a lateral axis of the trailer **140**; and defines a second sensor receptacle extending parallel to a longitudinal axis of the trailer **140**. Further, a first sensor (e.g., a strain gauge) is arranged in the first sensor receptacle and configured to output a signal representing shear forces in the kingpin **110** parallel to the lateral axis and a second sensor (e.g., a strain gauge) is arranged in the second sensor receptacle and configured to output the second signal representing shear forces in the kingpin **110** parallel to the longitudinal axis.

(59) In yet another variation, the shank **116** defines: a first blind bore extending parallel to a lateral axis of the trailer **140**; and a second blind bore extending parallel to a longitudinal axis of the trailer **140** and laterally offset from the first blind bore. Further, a first sensor (e.g., a strain gauge) is arranged in the first blind bore configured to output a signal representing shear forces in the kingpin **110** parallel to the lateral axis and a second sensor (e.g., a strain gauge) is arranged in the second blind bore and configured to output a signal representing shear forces in the kingpin **110** parallel to the longitudinal axis. Thus, the first blind bore and the second blind bore cooperate to

achieve a target wall thickness of the shank **116** (e.g., reduce the wall thickness) and increase shear force sensitivity of the force sensors arranged within the first- and second-blind bores.

(60) 4.2 Sensors

(61) Generally, the kingpin **110** can include a set of sensors **119**—such as force sensors (e.g., a strain gauge, an IMU, a load cell), optical sensors (e.g., a one-dimensional depth sensor, a LIDAR sensor, an RGB camera), and/or inertial sensors (e.g., an IMU, an accelerometer, a gyroscope)—coupled to the base in, the shank **116**, and/or the head **117** of the kingpin **110**.

(62) Furthermore, each sensor in the suite of sensors **119** can generate sensor data (e.g., analog values, digital values) in a sense domain such as including: lateral forces applied to the kingpin **110**; longitudinal forces applied to the kingpin **110**; vertical forces applied to the kingpin **110**; locations of the trailer **140**; speeds of the trailer **140**; battery charge states; incline angles (e.g., grade of a road) of the trailer **140**; accelerations of the trailer **140**; and/or tractor-trailer **140** angle of the trailer **140**, etc. Each sensor can then transmit these sensor data to the controller **150**.

(63) In one example, a set of (e.g., two) sensors **119** are mounted to the base in of the kingpin **110**. The first sensor includes a first strain gauge extending laterally across a front face of the kingpin **110**; and configured to output a first signal corresponding to shear forces applied to the kingpin **110** parallel to a lateral axis of the trailer **140**. The second sensor in the set of sensors includes a second strain gauge extending longitudinally across a side face of the kingpin **110** and configured to output a second signal corresponding to shear forces applied to the kingpin **110** parallel to a longitudinal axis of the trailer **140**.

(64) In another example, a set of sensors **119** are mounted to the shank **116** of the kingpin **110**.

(65) 4.2.1 Force Sensors

(66) In one implementation, the kingpin **110** can include a set of force sensors **119**. In this implementation, the kingpin **110** can include: a first sensor **119** configured to output signals representing lateral forces (e.g., loads) applied to the kingpin **110**; and a second sensor **119** configured to output signals corresponding to longitudinal forces (e.g., loads), parallel to a longitudinal axis **143** of the trailer **140**, applied to the kingpin **110**.

(67) In one variation, the kingpin **110** can include a set of (e.g., two) force sensors mounted to the base in of the kingpin **110**. A first sensor, in the set of force sensors: includes a first strain gauge extending laterally across a front face of the kingpin **110**; and is configured to output a first signal corresponding to shear forces applied to the kingpin **110** parallel to a lateral axis of the trailer **140**. A second sensor, in the set of force sensors, includes a second strain gauge extending longitudinally across a side face of the kingpin **110**, adjacent the front face of the kingpin **110**, and configured to output a second signal corresponding to shear forces applied to the kingpin **110** parallel to a longitudinal axis of the trailer **140**.

(68) Additionally or alternatively, the kingpin **110** can include a third sensor **119** (e.g., a third strain gauge) configured to output signals representing vertical forces applied to the kingpin **110**. This sensor **119** can then transmit signals representing a vertical force applied to the kingpin **110** to the controller **150** to identify a coupling event and selectively alternate between a tow mode and a manual mode, as further described below.

(69) 4.2.2 Other Sensors

(70) In one implementation, the kingpin **110** can include additional sensors **119**, such as other force sensors (e.g., a load cell, an IMU), optical sensors (e.g., a one-dimensional depth sensor, a LIDAR sensor, an RGB camera), proximity sensors (e.g., a Hall effect sensor, a conductive sensor, an inductive sensor), and/or inertial sensors (e.g., an IMU, an accelerometer, a gyroscope) etc.

(71) In one variation, the kingpin **110** can include an inertial measurement unit (e.g., an IMU) configured to output signals representing motion in pitch, roll, and yaw positions of the kingpin **110** and/or angular velocity of the trailer **140**. The inertial measurement unit can then transmit these signals to the controller **150**.

(72) In another variation, the kingpin **110** can include: a load cell configured to output signals

representing tension, compression, pressure, or torque applied to the kingpin **110** and transmit these signals to the controller **150**.

(73) In yet another variation, the kingpin **110** can include an accelerometer configured to output signals representing acceleration or speed of the trailer **140** at the kingpin **110**. The accelerometer can then transmit these signals to the controller **150**.

(74) 5. Interface

(75) As shown in FIGURES SA, 5B, 5C, and 5D, the system **100** can include an interface **120** including a joystick **121** and a kingpin **110** interface **123** configured to transiently couple to the kingpin **110** and to transfer forces (e.g., lateral forces, longitudinal forces, vertical forces), applied to the joystick **121**, into the kingpin **110**.

(76) In one implementation, the interface **120** includes the joystick **121** configured to move within a range of positions and the kingpin **110** interface **123** includes a clamp configured to transiently couple to the kingpin **110** and transfer forces applied to the joystick **121**. In this implementation, the joystick **121** is manually manipulated by a user within the range of positions or manipulated by a remote control **122** and the kingpin **110** interface **123** further includes a display configured to present forces applied to the joystick **121** to a user.

(77) In another implementation, the interface **120** includes an analog joystick (e.g., a control stick) configured to move within a range of positions and the analog joystick is manually manipulated by a user or manipulated by a remote control **122** within the range of positions. The interface **120** is configured to transiently couple to the kingpin **110** and transfer forces, applied to the analog joystick, to the kingpin **110**. However, the system **100** can include any other type of control stick and include any other type of kingpin **110** interface **123**.

(78) 6. Trailer+Drive System

(79) As shown in FIGS. 3A, 3B, 3C, and 8A, the trailer **140** includes: a set of rails **147**; a drive system **130** (e.g., a bogie); a motor **131**; a right wheel **145**; a left wheel **144**; and landing gear **141**.

(80) In one implementation, the trailer **140** includes: a floor **146**; a left rail **147** coupled to the floor **146**, extending parallel to and laterally offset from a longitudinal centerline **143** of the trailer **140**, and defining a first array of engagement features distributed along the left rail and longitudinally offset by a pitch distance; a right rail **147** coupled to the floor **146**, extending parallel to and laterally offset from the longitudinal centerline **143** of the trailer **140** opposite the left rail **147**, and defining a second array of engagement features distributed along the right rail **147** and longitudinally offset by the pitch distance; and a drive system **130**.

(81) In this implementation, the drive system **130** includes: a chassis; a set of latches configured to transiently engage a subset of engagement features, in the first array of engagement features on the left rail **147** and in the second array of engagement features in the right rail **147**, to retain the chassis on a distal end of the trailer **140** opposite the kingpin **110**; a passive axle **134** including a left passive wheel **135** and a right passive wheel **136**; a driven axle **137** adjacent the passive axle **134** and including a left driven wheel **138** and a right driven wheel **139**; and a motor **131** mounted to the chassis, coupled to the drive axle **137**; and configured to output torque to the left driven wheel **138** and to the right driven wheel **139**. The controller **150** is configured to trigger the motor **131** coupled to the left driven wheel **138** and the right driven wheel **139** to output torque in the direction of a force applied to the kingpin **110**.

(82) In one variation, the drive system **130** includes a primary motor **131** coupled to the wheels of the driven axle **137**. For example, the drive system **130** is arranged on a floor of the trailer **140** proximal the distal end **149** of the trailer **140** opposite the kingpin **110** and includes: a passive axle **134** and a driven axle **137** adjacent the passive axle **134**. The passive axle **134** includes: a left passive wheel **135** and a right passive wheel **136**. The driven axle **137** includes: a left driven wheel **138**; a right driven wheel **139**; and a primary motor **131** coupled to the left driven wheel **138** and the right driven wheel **139**. The drive system **130** is configured to trigger the primary motor **131** to drive the left driven wheel **138** and the right driven wheel **139** and thus, output torque in a tow

mode, as shown in FIGS. 8B and 8C.

(83) Furthermore, the system **100** can further include a battery **106** coupled to the floor **146** of the trailer **140** and electrically coupled to the drive system **130**. In one variation, the trailer **140** includes a set of battery packs, configured to form the battery **106**, coupled to the floor **146** of the trailer **140** and electrically coupled to the drive system **130**. In this variation, the controller **150** is configured to detect a charge state (e.g., a status, a level, a percentage) of each battery pack in the battery **106** and calculate a target preload force inversely proportional to the charge state of each battery pack, as further described below.

(84) 6.1 Landing Gear

(85) In one implementation, shown in FIG. 5B, the trailer **140** includes a landing gear **141** configured to support the trailer **140**. The landing gear **141** includes: a leg coupled to the floor **146** of the trailer **140** along the longitudinal axis of the trailer **140**; and a set of wheels **142** (e.g., caster wheels) pivotably coupled to the leg and configured to engage a ground surface below the floor **146** of the trailer **140**. The trailer **140** can further include a wheel actuator arranged on the landing gear **141** and configured to transition the set of wheels **142** from a retracted position (e.g., proximal a distal end of the leg) to an extended position (e.g., engaging a ground surface below the floor **146** of the trailer **140**) and thereby, enable motion of the trailer **140** in a manual mode in conjunction with the left wheel of the trailer **140**, the right wheel **145** of the trailer **140**, the left driven wheel **138**, and the right driven wheel **139**.

(86) Therefore, the controller **150** can trigger the wheel actuator to extend the set of wheels of the landing gear **141** toward a ground surface in a manual mode and retract the set of wheels toward the distal end of the leg of the landing gear **141** at the end of a manual mode and enter a tow mode.

(87) 6.2 Other Sensors

(88) In one implementation, the system **100** can include additional sensors **119**, such as other force sensors (e.g., a load cell, an IMU), optical sensors (e.g., a one-dimensional depth sensor, a LIDAR sensor, an RGB camera), proximity sensors (e.g., capacitive sensors, inductive sensors, Hall effect sensors), amplifier sensors, pressure sensors, and/or inertial sensors (e.g., an IMU, an accelerometer, a gyroscope) arranged on the trailer **140**.

(89) In one variation, an accelerometer is arranged on the trailer **140** proximal the kingpin **110** and is configured to output signals representing accelerations of the trailer **140**. The accelerometer can then transmit these signals to the controller **150**.

(90) In another variation, a set of distance sensors are arranged on the trailer **140**. In this variation, a left distance sensor is arranged on the trailer **140** proximal the left side of the kingpin **110** and is configured to output a first distance signal representing a first distance between the trailer **140** and a left feature on a corresponding left side of the tow vehicle. Further, a right distance sensor is arranged on the trailer **140** proximal the right side of the kingpin **110** and configured to output a second distance signal representing a second distance between the trailer **140** and a right feature on the right side of the tow vehicle. The controller **150** can then access the first distance signal from the left distance sensor and the second distance signal from the right distance sensor; calculate a tractor-trailer **140** angle based on the first distance signal and the second distance signal; and trigger the motor to output torque in response to correspondence between a direction of a force applied to the kingpin **110** and the tractor-trailer **140** angle, as further described below.

(91) 7. Controller

(92) The controller **150** is coupled to sensors **119** within the system **100** and executes methods and techniques described below to: calculate a direction and a magnitude of each force applied to the kingpin **110**; identify a coupling and/or a decoupling event between a hitch (e.g., a fifth wheel) of a tow vehicle (e.g., a tractor-trailer) and the kingpin **110** based on these forces; trigger a primary motor **131** and a primary airbrake system **107** to manipulate driven wheels of the trailer **140** in a manual mode; calculate a target preload force as a function of a condition of the trailer **140** (e.g., a speed, an incline angle, a tractor-trailer angle, a location, a charge state of a battery) in a tow mode;

trigger the primary motor **131**, arranged in the drive system, to selectively reduce torque output and/or increase torque output to decrease a difference between each force and the target preload force in the tow mode; and identify errors at the kingpin **110** (e.g., abusive movement, sensor **119** failure, unauthorized user).

(93) In one example, the controller **150** implements a Kalman filter to calculate the target preload force based on signals output from the sensors and triggers the primary motor **131** to selectively reduce torque output and/or increase torque output to decrease a difference between each force and the target preload force in the tow mode.

(94) 8. Kingpin Setup Period

(95) Generally, the system **100** can include a lockout system (e.g., a mechanical propulsion lockout system, an electrical lockout system) to prevent an unauthorized user, device, or vehicle from accessing the kingpin **110**, the trailer **140**, and/or any data (e.g., force data, voltage data, controller area network data) associated with the trailer **140**.

(96) In one implementation, as shown in FIG. 7A, during the setup period, the controller **150** can: enter a hibernation mode; detect a coupling event between the kingpin **110** and a tow vehicle in Block **S116**; and, in response to detecting the coupling event between the kingpin **110** and the tow vehicle, trigger the wireless communication module arranged on the trailer **140** to transmit wireless interrogation signals to nearby devices (e.g., within a threshold distance of the trailer **140**) in Block **S118**. Then, the controller **150** can access an identifier broadcast by a device and, in response to detecting the identifier in a database of identifiers of approved devices, enter a tow mode.

Alternatively, in response to detecting absence of the identifier in the database, the controller **150** can enter a hibernation mode and disable torque output by the motor **131** in hibernation mode or enter an antitheft mode and trigger the motor **131** to output a braking torque to prevent unauthorized access to the kingpin **110** and the trailer **140** in Block **S150**.

(97) In one variation, a user (e.g., a driver, an operator) may execute a sequence of steps or manually shift an external switch on the trailer **140** from a closed position to an open position and the controller **150** can then deactivate a hibernation mode and enter a tow mode.

(98) 8.1 Wired Connection Lockout: Hibernation Mode to Tow Mode

(99) In one implementation, during the setup period, a user (e.g., a driver, an operator) may manually execute a sequence of steps or shift an external switch on the trailer **140** from a closed position to an open position, and the controller **150** can then deactivate the hibernation mode and enter a tow mode.

(100) In one variation, a driver may couple a hitch of a tow vehicle to the kingpin **110** and then execute a particular sequence of steps such as applying the brakes of the tow vehicle for a duration of time, turn the steering wheel of the tow vehicle in a particular direction, and/or press a series of buttons within the tow vehicle. Then, in response to detecting the particular sequence of steps, the controller **150** can: deactivate the hibernation mode; and enter a tow mode.

(101) In another variation, the trailer **140** can include an external switch on the proximal end **148** of the trailer **140** and accessible by an operator of the trailer **140**. The operator may then manually shift the external switch from a closed position to an open position. Then, the controller **150** can: detect a force applied to the kingpin **110** by the hitch of a tow vehicle via a force sensor; and, in response to detecting the force applied to the kingpin **110**, identify the force as a coupling event with the tow vehicle, deactivate the hibernation mode, and enter a tow mode.

(102) In yet another variation, the battery is electrically coupled to a charge station (e.g., via a power cable) and the operator may manually decouple the battery from the charge station by disconnecting the power cable from the battery. Then, the controller **150** can: detect a force applied to the kingpin **110** by the hitch of a tow vehicle via a force sensor; and, in response to detecting the force applied to the kingpin **110**, identify the force as a coupling event with the tow vehicle, deactivate the hibernation mode, and enter a tow mode.

(103) Therefore, a user (e.g., a driver, an operator) may manually execute a particular sequence of

steps, shift an external switch of the trailer **140**, and/or manually decouple the battery from a charge station during the setup period and the controller **150** can deactivate the hibernation mode and enter a tow mode.

(104) 8.2 Wireless Connection Lockout: Hibernation Mode to Tow Mode

(105) In one implementation, in Block **S118**, in response to detecting a coupling event between the kingpin **110** and the hitch of a tow vehicle, the controller **150** can trigger the wireless communication module arranged on the trailer **140** to transmit wireless interrogation signals to nearby devices (e.g., within a threshold distance of the trailer **140**). Then, the controller **150** can access an identifier broadcast by a device and, in response to detecting the identifier in a database of identifiers of approved devices, enter a tow mode, as shown in FIG. 7A.

(106) In one variation, the trailer **140** can include a radio frequency identification tag (e.g., an RFID tag) or a near-field communication chip (e.g., an NFC chip) and a user may access a device (e.g., a mobile phone, a tablet, a computing device) configured to read a particular radio frequency associated with the RFID tag or the NFC chip on the trailer **140** and locate the device within a threshold distance of the trailer **140**. The device can then implement a handshake protocol with the wireless communication module on the trailer **140** and enter a tow mode.

(107) For example, during the setup period, a driver of a tow vehicle may access a mobile phone configured to read data stored on RFID tags and locate the mobile phone within a threshold distance (e.g., three feet) of an RFID tag on the trailer **140**. The mobile phone can then implement a handshake protocol with the RFID tag on the trailer **140** and, in response to detecting an identifier from the mobile phone associated with the RFID tag, the controller **150** can enter a tow mode.

(108) Furthermore, the controller **150** can repeat methods and techniques described above to identify a decoupling event with a tow vehicle, deactivate the tow mode, and enter the hibernation mode.

(109) Therefore, the controller **150** can access identifiers broadcast by devices proximal the trailer **140** and leverage a database of identifiers of approved devices to deactivate the hibernation mode and enter a tow mode.

(110) 8.3 Wireless Connection Lockout: Antitheft Mode

(111) In one variation, as shown in FIG. 7A, the trailer **140** can include a wireless communication module: arranged proximal the kingpin **110**; configured to broadcast wireless interrogation signals to devices within a threshold distance of the trailer **140**; and receive identifiers from these devices. The controller **150** can then: access a database of identifiers of approved devices; verify presence of these identifiers within the database; and enter a tow mode. Alternatively, the controller **150** can: verify absence of these identifiers within the database; enter an antitheft mode; and trigger the motor **131** to output a braking torque to prevent unauthorized access to the kingpin **110** and the trailer **140** in Block **S150**.

(112) For example, during the setup period, the controller **150** can: detect a coupling event between a tow vehicle and the kingpin **110** based on an initial signal from the first sensor **119**; and, in response to detecting the coupling event between the tow vehicle and the kingpin **110**, trigger the wireless communication module to transmit a wireless interrogation signal to devices within a threshold distance of the trailer **140**. The controller **150** can then access an identifier, broadcast by a device (e.g., a tablet), received by the wireless communication module responsive to the wireless interrogation signal. In response to detecting absence of the identifier in the database of identifiers of approved devices, the controller **150** can: enter an antitheft mode; access a first signal from the first sensor representing a lateral force applied to the kingpin **110**; access a second signal from the second sensor representing a longitudinal force applied to the kingpin **110**; calculate a direction and a magnitude of a force applied to the kingpin **110** by the hitch of the tow vehicle based on the first signal and the second signal; and trigger the motor to output a braking torque opposite the direction and proportional to the magnitude.

(113) Therefore, the controller **150** can access identifiers broadcast by devices proximal the trailer

140 and leverage a database of identifiers of approved devices to enter an antitheft mode and trigger the motor to output a braking torque and thereby, prevent unauthorized access, modification, movement, or theft of the kingpin **110** and/or the trailer **140**.

(114) 9. Manual Mode

(115) Generally, the controller **150** can identify a decoupling event between the kingpin **110** and a tow vehicle and deactivate the tow mode. Then, a user or a machine can couple the interface **120** to the shank **116** of the kingpin **110** and the controller **150** can enter a manual mode (e.g., tow vehicle less control of the trailer **140**). In manual mode, the drive system **130** can trigger the primary motor **131** to drive the left wheel **144** and the right wheel **145** of the trailer **140** according to target wheel speeds to locate the trailer **140** in a target position in Block **S190**.

(116) Furthermore, at the start of the manual mode, the controller **150** can trigger the wheel actuator to extend the set of wheels of the landing gear **141** toward a ground surface below the floor **146** of the trailer **140**. The controller **150** can then: access a signal from a set of force sensors **119**; detect a null vertical force applied to the kingpin **110** based on the signal; and activate a manual mode in response to detecting the null vertical force. The controller **150** can calculate a maximum wheel speed for a left wheel and a right wheel of the trailer **140** and thereby, limit the swing radius (e.g., pivot radius) of the trailer **140** in manual mode. At the end of manual mode, the controller **150** can trigger the wheel actuator to retract the set of wheels of the landing gear **141** toward a distal end of a leg of the landing gear **141**.

(117) In one implementation, shown in FIGS. **10** and **7A**, the controller **150** can: access a signal from the set of sensors **119** in Block **S170**; based on the signal, calculate a first direction and a first magnitude of a first force applied to the kingpin **110** via the joystick **121** by a user or a remote control **122** communicatively coupled to the joystick **121** in Block **S170**; calculate a target left wheel speed of a left wheel **144** of the trailer **140** proportional to the first magnitude in Block **S174**; calculate a target right wheel speed of a right wheel **145** of the trailer **140** proportional to the first magnitude, the target right wheel speed differing from the target left wheel speed based on the first direction in Block **S176**; and serve the target left wheel speed and the target right wheel speed to the drive system **130** arranged proximal a distal end **149** of the trailer **140** opposite the kingpin **110** in Block **S178**. The drive system **130** can then trigger the motor **131** to drive a left driven wheel **138** and a right driven wheel **139** of the trailer **140** in Block **S190**.

(118) 9.1 Wireless Connection Lockout: Hibernation Mode to Manual Mode

(119) In one variation, the interface **120** can include a wireless transmitter configured to broadcast an identifier (e.g., a barcode, a number, a digital key) to a wireless receiver arranged on the proximal end of the trailer **140** and configured to receive identifiers from the interface **120**. The controller **150** can then verify receipt of the identifier to selectively enter a manual mode and/or a tow mode.

(120) For example, during the setup period, the interface **120** can broadcast a first identifier (e.g., 126374958). The controller **150** can then: access the identifier (e.g., 126374958) from the interface **120** via the wireless receiver; and, in response to verifying receipt of the identifier, enter a tow mode to calculate a maximum wheel speed. Once the controller **150** calculates the maximum wheel speed, the interface **120** can broadcast a second identifier (e.g., 56123379). The controller **150** can then: access the second identifier (e.g., 56123379) from the interface **120** via the wireless receiver; and, in response to verifying receipt of the identifier, enter a manual mode. Thus, the controller **150** can verify receipt of identifiers to selectively enter a manual mode and/or a tow mode.

(121) 9.1.1 Maximum Wheel Speed

(122) In one variation, upon verification of receipt of an identifier, the controller **150** can: enter a tow mode; pulse the motor **131** to rotate the right wheel **145** and the left wheel **144** of the trailer **140**; and access an acceleration of the trailer **140** at the kingpin **110** (e.g., via an accelerometer). The controller **150** can then leverage the acceleration to calculate a maximum wheel speed for the left wheel **144** and the right wheel **145** of the trailer **140** and thereby, limit the swing radius (e.g.,

pivot radius) of the trailer **140** in manual mode.

(123) For example, during the setup period, in response to verifying receipt of the identifier, the controller **150** can enter a tow mode. In tow mode, the controller **150** can: pulse the primary motor **131** to rotate the left wheel **144** and the right wheel **145** of the trailer **140** by a rotation distance less than ten degrees of a full rotation; access a signal representing an initial acceleration of the trailer **140** at the kingpin **110** from an inertial sensor **119** (e.g., an accelerometer); and interpret a distance between the kingpin **110** and the drive system **130** based on the signal. The controller **150** can then calculate a maximum wheel speed proportional to the distance and implement methods and techniques described above to enter a manual mode. Once the kingpin **110** is in manual mode, the controller **150** can: calculate the target left wheel speed of the left wheel **144** of the trailer **140** less than the maximum wheel speed; and calculate the target right wheel speed of the right wheel **145** of the trailer **140** less than the maximum wheel speed.

(124) Additionally or alternatively, the controller **150** can implement a short-range wireless communication protocol to: broadcast reference signals from a transmitter in the interface to a receiver arranged on the drive system **130**; and record transmit and receive durations (or “time of arrival receipts”) for these reference signals. The controller **150** can then transform a reference signal into a time-based distance between the interface **120** and the drive system **130** and calculate the maximum wheel speed proportional to this time-based distance.

(125) For example, the interface **120** can include a transmitter configured to transmit a reference signal to the drive system **130**, the drive system **130** including a receiver arranged on the drive system **130** and configured to receive the reference signal from the transmitter. Then, during the setup period, the controller **150** can: record a time of arrival receipt of the reference signal arriving at the receiver arranged on the drive system **130**; transform the time of arrival receipt into a distance between the kingpin **110** and the drive system **130**; and calculate a maximum wheel speed proportional to the distance. The controller **150** can then implement methods and techniques described above to enter a manual mode. Once the kingpin **110** is in manual mode, the controller **150** can: calculate a target left wheel speed of the left wheel **144** of the trailer **140** less than the maximum wheel speed; and calculate a target right wheel speed of the right wheel **145** of the trailer **140** less than the maximum wheel speed.

(126) Therefore, the controller **150** can calculate a maximum wheel speed for the left wheel **144** and the right wheel **145** of the trailer **140** in a tow mode. Additionally, the controller **150** can enter a manual mode to set the maximum wheel speed for the left wheel **144** and the right wheel **145** of the trailer **140** to limit the swing radius of the trailer **140** in manual mode and thus, prevent the trailer **140** from colliding with external objects and the operator of the joystick **121**.

(127) 9.2 Manual Mode: Manual Joystick+Remote Control

(128) In one variation, the drive system **130** includes a primary motor **131** coupled to the wheels of the driven axle **137**. In this variation, the drive system **130** is arranged on a floor of the trailer **140** **146** proximal the distal end **149** of the trailer **140** opposite the kingpin **110** and includes a passive axle **134** and a driven axle **137** adjacent the passive axle **134**. The passive axle **134** includes a left passive wheel **135** and a right passive wheel **136**. The driven axle **137** includes: a left driven wheel **138**; a right driven wheel **139**; and a primary motor **131** coupled to the left driven wheel **138** and the right driven wheel **139**. The drive system **130** can trigger the primary motor **131** to drive the left wheel **144** and the right wheel **145** of the trailer **140** according to target wheel speeds in manual mode.

(129) For example, a user (e.g., a driver, an operator, a yard manager) may wish to relocate a trailer **140** parked in a yard to a new position. Accordingly, the user may: access and clamp (e.g., attach) the interface **120** onto the shank **116** of the kingpin **110** of this trailer **140** parked in the yard and manipulate the joystick **121** in a first direction toward the new position. The kingpin **110** interface **123** can then identify the direction of the joystick **121** as a lateral force applied to the joystick **121** and transfer this lateral force into the shank **116**. A force sensor **119** coupled to the shank **116** can

then output a signal representing this lateral force and the controller **150** can: calculate a first direction and a first magnitude of the force applied to the kingpin **110**; calculate a target left wheel speed of a left wheel **144** of the trailer **140** proportional to the first magnitude; calculate a target right wheel speed of the right wheel **145** of the trailer **140** proportional to the first magnitude, the target right wheel speed differing from the target left wheel speed based on the first direction; and serve these target wheel speeds to the drive system **130**. The drive system **130** can then: trigger the left motor to drive the left driven wheel **138** according to the target left wheel speed; and trigger the right motor to drive the right driven wheel **139** according to the target right wheel speed to relocate the trailer **140** to the new position.

(130) Alternatively, the user (e.g., a driver, an operator, a yard manager) may access a remote control **122** communicatively coupled to the interface **120**. The user may then set a target position for the trailer **140** at the remote control **122** and the remote control **122** can transmit the target position to the interface **120**. The interface **120** can then derive a force from the target position and transfer this force into the shank **116**. The controller **150** and the drive system **130** can then implement methods and techniques described above to relocate the trailer **140** to the target position.

(131) Therefore, in manual mode, the user may manually manipulate the joystick **121** and the drive system **130** can drive the wheels of the trailer **140** to relocate the trailer **140** according to forces applied to the joystick **121** by the user. Additionally, the user may access a remote control **122** and set a target position for the trailer **140** and the drive system **130** can drive the wheels of the trailer **140** to relocate the trailer **140** to the target position according to forces transferred to the joystick **121** by the remote control **122**.

(132) 9.3 Manual Mode: Drive System Variations

(133) In one variation, shown in FIG. 8D, the drive system **130** includes a set of secondary motors **132** and each motor in the set of secondary motors **132** is coupled to a wheel of the driven axle **137**. In this variation, the drive system **130** is arranged on a floor of the trailer **140** proximal the distal end **149** of the trailer **140** opposite the kingpin **110** and includes: a passive axle **134** and a driven axle **137** adjacent the passive axle **134**. The passive axle **134** includes: a left passive wheel **135** and a right passive wheel **136**. The driven axle **137** includes: a left driven wheel **138**; a right driven wheel **139**; a left motor **132** coupled to the left driven wheel **138**; and a right motor **132** coupled to the right driven wheel **139**. The drive system **130** can: receive the target right wheel speed and the target left wheel speed from the controller **150**; trigger the left motor **132** to drive the left driven wheel **138** according to the target left wheel speed in Block **S192**; and trigger the right motor **132** to drive the right driven wheel **139** according to the target right wheel speed in Block **S194**.

(134) Additionally or alternatively, the drive system **130** includes a primary motor **131** coupled to the wheels of the driven axle **137** and the set of secondary motors **132**. The drive system **130** further includes: a primary motor **131** coupled to the left driven wheel **138** and the right driven wheel **139**; a left motor **132** coupled to the left driven wheel **138**; and a right motor **132** coupled to the right driven wheel **139**. The drive system **130** can then: receive the target right wheel speed and the target left wheel speed from the controller **150**; trigger the primary motor **131** to drive the left wheel and the right wheel according to the first direction and the first magnitude in Block **S190**; and selectively trigger the left motor **132** and the right motor **132** according to a difference between the target left wheel speed and the target right wheel speed in Block **S196**.

(135) In yet another variation, the drive system **130** includes a primary motor **131** coupled to the wheels of the driven axle **137** and a primary airbrake system **107** coupled to an airbrake supply of the tow vehicle. In this variation, the drive system **130** is arranged on the floor **146** of the trailer **140** proximal the distal end **149** of the trailer **140** opposite the kingpin **110** and includes a passive axle **134** and the driven axle **137** adjacent the passive axle **134**. The passive axle **134** includes: a left passive wheel **135**; a right passive wheel **136**; and a primary airbrake system **107** configured to couple to an airbrake supply of the tow vehicle. The driven axle **137** includes: a left driven wheel

138; a right driven wheel **139**; and a primary motor **131** coupled to the left driven wheel **138** and the right driven wheel **139**. The drive system **130** can then: receive the target right wheel speed and the target left wheel speed from the controller **150**; trigger the primary motor **131** to drive the left wheel and the right wheel according to the first direction and the first magnitude in Block **S190**; and selectively operate the primary airbrake system **107** to influence the airbrake supply of the tow vehicle to the trailer **140** according to a difference between the target left wheel speed and the target right wheel speed in Block **S198**, as shown in FIG. 7C.

(136) Alternatively, the drive system **130** can include a primary motor **131** coupled to the wheels of the driven axle **137**, the primary airbrake system **107** coupled to the airbrake supply of the tow vehicle, and a secondary brake system **133** coupled to each wheel of the driven axle **137**. The drive system **130** further includes: a secondary left brake system **133** coupled to the left wheel and isolated from the primary airbrake system **107**; and a secondary right brake system **133** coupled to the right wheel and isolated from the primary airbrake system **107**. The drive system **130** can then: receive the target right wheel speed and the target left wheel speed from the controller **150**; trigger the primary motor **131** to drive the left wheel and the right wheel according to the first direction and the first magnitude in Block **S190**; and selectively operate the secondary left brake system **133** and the secondary right brake system **133** according to a difference between the target left wheel speed and the target right wheel speed in Block **S198**, as shown in FIG. 7C. Therefore, the drive system **130** can trigger a primary motor **131**, a set of secondary motors **132**, a primary airbrake system **107**, and/or a set of secondary brake systems **133** to manipulate the driven wheels of the trailer **140** and to relocate the trailer **140** to a new position.

(137) 9.4 Manual Mode: Joystick Control

(138) In one variation, the controller **150** can access signals from an accelerometer to detect accelerations of the trailer **140** in manual mode and leverage a maximum acceleration (e.g., a threshold acceleration) of the trailer **140** to set the target left wheel speed and the target right wheel speed of the trailer **140**. Further, at accelerations greater than the threshold acceleration, the controller **150** can set the target wheel speeds to null (e.g., zero) and the drive system **130** can trigger the motor **131** to slow the motion of the trailer **140** and thereby, prevent a user from losing control of the joystick **121**.

(139) For example, the controller **150** can: access a signal from an accelerometer proximal the kingpin **110** and representing an acceleration of the trailer **140**; interpret an acceleration of the trailer **140** at the kingpin **110** based on the signal; in response to the acceleration of the trailer **140** exceeding a threshold acceleration, set the target left wheel speed and the target right wheel speed to null; and serve the target left wheel speed and the target right wheel speed to the drive system **130**. The drive system **130** can then trigger the primary motor **131** to drive the left wheel according to the null target left wheel speed and the right wheel according to the null target right wheel speed to slow the motion of the trailer **140**.

(140) Thus, the controller **150** and the drive system **130** can cooperate to set target wheel speeds and drive the wheels according to these target wheel speeds to enable a user to maintain control of the joystick **121** in manual mode.

(141) 9.5 Manual Mode to Tow Mode

(142) At the end of manual mode, the controller **150** can interpret a vertical force applied to the kingpin **110** and identify a coupling event at the kingpin **110** with the hitch of a tow vehicle. The controller **150** can then deactivate the manual mode and activate a tow mode. The controller **150** can then trigger the wheel actuator to transition the set of wheels of the landing gear **141** from the extended position toward the floor of the trailer **140** **146** to a retracted position and enter a tow mode.

(143) 10. Tow Mode

(144) Generally, the user (e.g., an operator, a driver, a yard manager) or a machine (e.g., a forklift) couples the hitch (e.g., a fifth wheel) of a tow vehicle to the kingpin **110** and the controller **150** can

identify a coupling event between the kingpin **110** (e.g., via a signal from a force sensor) and the hitch of the tow vehicle. In particular, the controller **150** can: detect an initial force impulse applied to the kingpin **110**; interpret the initial force impulse as a coupling event with the hitch of the tow vehicle; and, in response to interpreting the initial force impulse as the coupling event with the hitch of the tow vehicle, enter a tow mode.

(145) In one implementation, in tow mode, the controller **150** can detect conditions of the trailer **140** such as including: a direction of motion of the trailer **140**; a tractor-trailer **140** angle (e.g., a steering angle); a speed of the trailer **140**; an incline angle of the trailer **140** (e.g., a grade of a ground surface); a location of the trailer **140**; forces applied to the kingpin **110** (e.g., lateral forces, longitudinal forces, vertical forces, total forces); and a charge state of each battery pack coupled to the trailer **140** etc. The controller **150** can then: calculate a target preload force proportional to and/or inversely proportional to the condition of the trailer **140**; and trigger a motor, arranged in the drive system, to increase torque output and/or reduce torque output in the direction of motion of the trailer **140** to decrease a difference between the target preload force and a total force applied to the kingpin **110** (e.g., an actual force) to control the trailer **140** in conjunction with the tow vehicle.

(146) More specifically, during a given time period (e.g., 30 seconds, five minutes, ten minutes, one hour), the controller **150** can: detect a direction of motion of the trailer **140**; set a target preload force opposite the direction of motion; detect a speed of the trailer **140**; and, in response to the speed exceeding a lower speed limit (e.g., 20 miles per hour, 30 miles per hour) enable torque output and regenerative braking at the drive system **130**. Then, the controller **150** can: calculate a speed variance over the time window based on the speed of the trailer **140**; adjust target torque output inversely proportional to the speed variance; and adjust the target preload force proportional to the speed variance. The controller **150** can further: detect a location of the trailer **140**; retrieve a georeferenced location or a low-emissions region from a drive route assigned to the trailer **140**; and, in response to the location falling within the threshold distance of the low-emissions region, increase target torque output and decrease the target preload force. Alternatively, in response to the location falling outside of a threshold distance of the low-emissions region, the controller **150** can decrease target torque output and increase the target preload force.

(147) Additionally, the controller **150** can: detect an incline angle of the trailer **140**; adjust target torque output proportional to the incline angle of the trailer **140**; and adjust target preload force inversely proportional to the incline angle of the trailer **140**. Similarly, the controller can: detect a decline angle of the trailer **140**; adjust target torque output inversely proportional to the decline angle of the trailer **140**; and adjust the target preload force inversely proportional to the decline angle. The controller **150** can: detect a charge state of a battery pack or the battery coupled to the trailer **140**; adjust target torque output proportional to the charge state of the battery pack or the battery; adjust target regenerative braking inversely proportional the charge state of the battery pack or the battery; and adjust the target preload force inversely proportional the charge state of the battery pack or the battery.

(148) Furthermore, the controller **150** can: detect a longitudinal force applied to the kingpin **110** by a hitch of a tow vehicle (e.g., via a first strain gauge); detect a lateral force applied to the kingpin **110** by the hitch (e.g., via a second strain gauge); calculate a tractor-trailer **140** angle (e.g., a steering angle) based on the longitudinal force and the lateral force; adjust target torque output inversely proportional the tractor-trailer **140** angle; and adjust the target preload force proportional the tractor-trailer **140** angle. The controller **150** can then: calculate a total force applied to the kingpin **110** by the hitch based on the lateral force and the longitudinal force. Accordingly, in response to the total force exceeding the target preload force, the controller **150** can trigger a motor **131**, arranged in the drive system **130**, to increase torque output in the direction of motion of the trailer **140** to decrease a difference between the target preload force and a total force applied to the kingpin **110** (e.g., an actual force) to control the trailer **140** in conjunction with the tow vehicle. Alternatively, in response to the total force falling below the target preload force, the controller **150**

can trigger the motor **131** to reduce torque output in the direction of motion of the trailer **140** to decrease a difference between the target preload force and the total force applied to the kingpin **110** (e.g., an actual force) to control the trailer **140** in conjunction with the tow vehicle.

(149) Therefore, the controller **150** can leverage the dynamic target preload force, torque output, and regenerative braking: to prevent a jackknife event between the tow vehicle and the trailer **140** (e.g., tow vehicle brakes to a stop and the trailer **140** continues to move toward and push the tow vehicle or torque applied to the kingpin **110**); to reduce emissions by the tow vehicle; to decrease fuel consumption by the tow vehicle; to extend the life of each battery pack coupled to the trailer **140**; and to enable a driver of the tow vehicle to apply minimal to no brake force to the trailer **140** in tow mode.

(150) At the end of tow mode, the controller **150** can identify a decoupling event between the kingpin **110** with the hitch of a tow vehicle and transition the kingpin **110** from the tow mode to the manual mode or from the tow mode to the hibernation mode.

(151) 10.1 Dynamic Target Preload Force: Tractor-Trailer Angle

(152) Blocks of the method **S100** recite: detect a direction of motion and a speed of the trailer **140** in Block **S120**; in response to the first total force falling below the first target preload force, triggering a motor **131**, arranged in a drive system **130** located proximal a distal end of the trailer **140**, to reduce torque output in the direction of motion to decrease a difference between the total force and the target preload force in Block **S150**; and, in response to a second total force falling below a second target preload force, triggering the motor **131** to increase torque output opposite the direction of motion to decrease a second difference between the second force and the second target preload force in Block **S160**.

(153) In one implementation, shown in FIGS. **4** and **7B**, the controller **150** can detect a direction of motion and a speed of the trailer **140**, detect a lateral force and a longitudinal force applied to the kingpin **110** by the hitch, and calculate a tractor-trailer **140** angle as a condition of the trailer **140**. Additionally, the controller **150** can access a lower speed limit (e.g., 25 miles per hour) and an upper speed limit of the trailer **140** (e.g., 65 miles per hour). The controller **150** can then leverage the tractor-trailer angle, the lower speed limit, and the upper speed limit to selectively increase, decrease, or disable torque output and adjust the target preload force.

(154) For example, the controller **150** can: detect a longitudinal force applied to a kingpin **110**, arranged on a proximal end of a trailer **140**, by a hitch of a tow vehicle (e.g., via a first strain gauge); detect a lateral force applied to the kingpin **110** by the hitch (e.g., via a second strain gauge); detect a speed and a direction of motion of the trailer **140** (e.g., 30 miles per hour in forward direction via an inertial measurement unit); based on the longitudinal force and the lateral force, calculate a tractor-trailer **140** angle, between longitudinal axes of the trailer **140** and the tow vehicle and calculate a total force, applied to the kingpin **110** by the hitch; calculate a target preload force opposite the first direction of motion (e.g., reverse direction) and proportional to the first tractor-trailer **140** angle; trigger a motor **131**, arranged in the drive system **130** located proximal a distal end of the trailer **140**, to reduce torque output in the direction of motion (e.g., forward direction) in order to decrease a difference between the total force and the target preload force, in response to the speed of the trailer **140** (e.g., 30 miles per hour) exceeding a threshold speed (e.g., a lower speed limit of 25 miles per hour) and in response to the total force falling below the target preload force.

(155) During a second time period, the controller **150** can: detect a second longitudinal force applied to the kingpin **110** by the hitch; detect a second lateral force applied to the kingpin **110** by the hitch; detect a second speed and a second direction (e.g., forward direction) of the trailer **140** (e.g., via an inertial measurement unit); based on the second longitudinal force and the second lateral force, calculate a second total force applied to the kingpin **110** by the hitch and calculate a second target preload force opposite the second direction of motion (e.g., a reverse direction) and proportional to the second tractor-trailer **140** angle; and trigger the motor **131** to increase torque

output in the second direction of motion (e.g., forward direction) to decrease a difference between the second total force and the second target preload force, in response to the second speed exceeding the threshold speed (e.g., a lower speed limit of 25 miles per hour) and in response to the second total force exceeding the second target preload force.

(156) During a third time period, the controller **150** can: detect a third longitudinal force applied to the kingpin **110** by the hitch; detect a third lateral force applied to the kingpin **110** by the hitch; detect a third speed and a third direction of motion of the trailer **140** (e.g., a forward direction); calculate a third total force applied to the kingpin **110** by the hitch of the tow vehicle based on the third longitudinal force and the third lateral force; and, in response to the third speed of the trailer **140** (e.g., 50 miles per hour) falling below a second threshold speed (e.g., an upper speed limit of 65 miles per hour), trigger the motor **131** to disable torque output in the third direction of motion. During a fourth time period, the controller **150** can: detect a fourth longitudinal force applied to the kingpin **110** by the hitch; detect a fourth lateral force applied to the kingpin **110** by the hitch; detect a fourth speed and a fourth direction of motion of the trailer **140** (e.g., 20 miles per hour in forward direction); in response to the fourth speed of the trailer **140** falling below the lower speed limit (e.g., 25 miles per hour), accessing a nominal low-speed preload force; calculating a fourth total force applied to the kingpin **110** by the hitch based on the fourth longitudinal force and the fourth lateral force; and, in response to the fourth total force exceeding the nominal low-speed preload force, trigger the motor **131** to increase torque output in the fourth direction of motion (e.g., forward direction) of the tow vehicle to decrease a difference between the fourth total force and the nominal low-speed preload force.

(157) 10.1.1 Dynamic Target Preload Force: Speed Variance+Duration

(158) In one variation, in Block **S122**, the controller **150** can calculate a speed variance for the first time period and for the second time period and leverage the tractor-trailer **140** angles, in the aforementioned example, and these speed variances to calculate the target preload force.

(159) For example, the controller **150** can: calculate a first speed variance of the trailer **140** during the first time period based on the first speed; calculate the first target preload force proportional the first tractor-trailer **140** angle and proportional the first speed variance; calculate the second tractor-trailer **140** angle less than the first tractor-trailer **140** angle; calculate a second speed variance of the trailer **140** during the second time period based on the second speed, the second speed variance greater than the first speed variance; and calculate the second target preload force, approximating the first target preload force, proportional the second tractor-trailer **140** angle and proportional the second speed variance.

(160) In another variation, the controller **150** can calculate a duration of the trailer **140** exhibiting the first speed during the first time period and exhibiting the first speed during the next time period. The controller **150** can then calculate the target preload force inversely proportional these durations for the corresponding time period.

(161) For example, the controller **150** can execute Blocks of the method **S100** to detect a first speed and a first direction of motion of the trailer **140** (e.g., 30 miles per hour in forward direction via an inertial measurement unit) and detect a first total force applied to the kingpin **110** by the hitch during a first time period. The controller **150** can then: detect a first duration of the trailer **140** exhibiting the first speed; and calculate a first target preload force opposite the direction of motion of the tow vehicle and inversely proportional the first duration. During a second time period, the controller **150** can: detect a second speed of the trailer **140** corresponding to the first speed of the trailer **140**; detect a second duration of the trailer **140** exhibiting the second speed; and, in response to the second duration exceeding the first duration, calculate a second target preload force, less than the first target preload force, opposite the direction of motion and inversely proportional the second duration. The controller **150** can then trigger the motor **131** to reduce torque output in the first direction of motion (e.g., forward direction) to decrease a first difference between the first total force and the first target preload force in response to the first total force falling below the first

target preload force.

(162) Therefore, the controller **150** can leverage the target preload force at the kingpin **110** to control the trailer **140** as the speed of the trailer **140** and a duration of the speed fluctuates during a drive route and thereby, enable a user (e.g., a driver) of the tow vehicle to apply minimal assistance to the trailer **140**.

(163) 10.1.2 Dynamic Target Preload Force: Stiction+Regenerative Braking

(164) In one variation, the controller **150** can detect a difference between the first total force and a second total force (e.g., a force impulse) for a duration of time while the trailer **140** is in motion. The controller **150** can then leverage the speed of the trailer **140**, a threshold difference, and a threshold duration of time to identify a stiction event between the kingpin **110** and the hitch of the tow vehicle.

(165) For example, the controller **150** can, during a first time period: detect a first total force applied to the kingpin **110** by the hitch of the tow vehicle; detect a speed and a direction of motion of the trailer **140** (e.g., 30 miles per hour in forward direction via an inertial measurement unit); based on the total force, calculate a tractor-trailer angle, between longitudinal axes of the trailer **140**; calculate a target preload force opposite the first direction of motion (e.g., reverse direction) and proportional to the first tractor-trailer angle; trigger a motor **131**, arranged in the drive system **130**, to reduce torque output in the direction of motion (e.g., forward direction) in order to decrease a difference between the total force and the target preload force, in response to the speed of the trailer **140** (e.g., 30 miles per hour) exceeding a threshold speed (e.g., a lower speed limit of 25 miles per hour) and in response to the total force falling below the target preload force.

(166) During a second period of time, the controller **150** can: detect a second total force applied to the kingpin **110** by the hitch of the tow vehicle; detect a speed and a direction of motion of the trailer **140** (e.g., 30 miles per hour in forward direction via an inertial measurement unit); detect a difference between the first total force and the second total force for a duration of time (e.g., 5 minutes); and, in response to the difference exceeding a threshold difference and in response to the duration exceeding a threshold duration of time, identify a stiction event between the kingpin **110** and the hitch of the tow vehicle and trigger the motor **131** to regeneratively brake the trailer **140**. The controller **150** can then detect a third total force applied to the kingpin **110**; and trigger the motor **131** to increase torque output in the direction of motion (e.g., forward direction) to decrease a difference between the third total force and the target preload force in response to the third total force exceeding the target preload force.

(167) Therefore, the controller **150** can monitor forces applied to the kingpin **110** to identify a stiction event between the kingpin **110** and the hitch of the tow vehicle and trigger the motor **131** to selectively apply a regenerative braking force to slow motion of the trailer **140**.

(168) 10.2 Dynamic Target Preload Force: Incline Angle of Trailer

(169) In one implementation, in Block S**134**, the controller **150** can detect an incline angle of the trailer **140** representing a grade of the road (e.g., 5% grade, 10% grade) as a condition of the trailer **140** and then leverage the incline angle to selectively adjust the target preload force. Additionally, in response to the incline angle exceeding a threshold incline angle (e.g., trailer **140** is moving up a hill), the controller **150** can decrease the target preload force. Alternatively, in response to the incline angle falling below the threshold incline angle (e.g., trailer **140** is moving down a hill), the controller **150** can increase the target preload force.

(170) For example, during a first time period, the controller **150** can: detect a first longitudinal force applied to the kingpin **110** by the hitch; detect a first lateral force applied to the kingpin **110** by the hitch of the tow vehicle; detect a first direction of motion of the trailer **140** (e.g., a forward direction); detect a first incline angle of the trailer **140** (e.g., 2.86 degrees, trailer **140** is moving up a hill); calculate a first total force, applied to the kingpin **110** by the hitch of the tow vehicle, based on the first longitudinal force and the first lateral force; calculate a first target preload force opposite the first direction of motion (e.g., a reverse direction) and proportional to the first incline

angle (e.g., decrease target preload force); and, in response to the first total force falling below the first target preload force, trigger the motor **131** to reduce torque output in the first direction of motion (e.g., forward direction) in order to decrease a first difference between the first total force and the first target preload force.

(171) During a second time period, the controller **150** can: detect a second force applied to the kingpin **110** by the hitch of the tow vehicle; detect a decline angle of the trailer **140** (e.g., -0.573 degrees, trailer **140** is moving down a hill); calculate a second target preload force opposite the direction of motion (e.g., a reverse direction) of the tow vehicle and inversely proportional the decline angle (e.g., increase target preload force); and, in response to the second force falling below the second target preload force, trigger the motor **131** to increase torque output opposite the direction of motion (e.g., reverse direction) in order to decrease the second difference between the second force and the second target preload force.

(172) During a third time period, the controller **150** can: detect a third total force applied to the kingpin **110** by the hitch of the tow vehicle; detect the decline angle of the trailer **140** (e.g., -0.573 degrees, trailer **140** is moving down a hill); calculate a third target preload force opposite the second direction of motion and proportional to the decline angle of the trailer **140** (e.g., decrease target preload force); and, in response to the second total force exceeding the second target preload force, trigger the motor **131** to regeneratively brake the driven axle to slow motion of the trailer **140** and to decrease a second difference between the second total force and the second target preload force.

(173) Therefore, the controller **150** can leverage the target preload force at the kingpin **110** to control the trailer **140** as the incline angle of the trailer **140** fluctuates as the trailer **140** navigates road grades during a drive route and thereby, reduce emissions and fuel consumption by the tow vehicle.

(174) 10.3 Dynamic Target Preload Force: Drive Route+Location of Trailer

(175) In one variation, in Block **S136**, the controller **150** can detect a location of the trailer **140** as a condition of the trailer **140** and then leverage the location and a drive route assigned to the trailer **140** to selectively increase or decrease the target preload force.

(176) Furthermore, an operator may define a start location and an end location for a drive route and upload this drive route to a user portal. The controller **150** can then: access the drive route; estimate a set of legs between the start location and the end location for the drive route; and populate each leg of the drive route with a time window, a corresponding georeferenced location, and emission conditions associated with the georeferenced location (e.g., a low-emissions region). The computer system can then access this drive route at the start of the tow mode. The controller **150** can also access signals from an inertial sensor to detect locations of the trailer **140** during the drive route.

(177) For example, during a first time period, the controller **150** can: detect a first longitudinal force applied to the kingpin **110** by the hitch of the tow vehicle; detect a first lateral force applied to the kingpin **110** by the hitch; detect a first direction of motion and first speed of the trailer **140** (e.g., a forward direction); detect a first tractor-trailer **140** angle; calculate a first total force, applied to the kingpin **110** by the hitch, based on the first longitudinal force and the first lateral force; calculate a first target preload force opposite the first direction of motion (e.g., a reverse direction) and proportional the tractor-trailer **140** angle; and, in response to the first total force falling below the first target preload force, trigger the motor **131** to reduce torque output in the first direction of motion (e.g., forward direction) in order to decrease a first difference between the first total force and the first target preload force.

(178) During a second time period, the controller **150** can: detect a second longitudinal force applied to the kingpin **110** by the hitch of the tow vehicle; detect a second lateral force applied to the kingpin **110** by the hitch; detect a second direction of motion and a second speed of the trailer **140** (e.g., a forward direction); detect a second tractor-trailer **140** angle; retrieve a first low-emissions region from a drive route assigned to the trailer **140**; detect a first location of the trailer

140 (e.g., via an inertial measurement unit); calculate a second total force, applied to the kingpin **110** by the hitch, based on the second longitudinal force and the second lateral force; calculate a second target preload force opposite the second direction of motion (e.g., a reverse direction) and proportional the second tractor-trailer **140** angle; and, in response to the first location falling within the threshold distance of the low-emissions region and in response to the second speed exceeding the first threshold speed (e.g., 25 miles per hour) and in response to the second total force exceeding the target preload force, trigger the motor **131** to reduce torque output in the second direction of motion (e.g., forward direction) in order to decrease a second difference between the first total force and the first target preload force.

(179) During a third time period, the controller **150** can: detect a third speed and a third direction of motion of the trailer **140** (e.g., forward direction); retrieve a last georeferenced location (e.g., a storage yard) from the drive route assigned to the trailer **140**; detect a second location of the trailer **140**; and, in response to the third speed of the trailer **140** falling below the first threshold speed and in response to the second location of the trailer **140** falling within the threshold distance of the last georeferenced location (e.g., a storage yard), trigger the motor to disable torque output in the third direction of motion (e.g., forward direction).

(180) Therefore, the controller **150** can selectively increase or decrease the target preload force and the target torque output by the motor as the location of the trailer **140** fluctuates during the drive route. Additionally, the controller **150** can also leverage the target preload force and the location of the trailer **140** to enable a user (e.g., a driver) of the tow vehicle to apply minimal to no brakes to the trailer **140**.

(181) 10.4 Dynamic Target Preload Force: Charge State of Battery

(182) In one variation, in Block **S138**, the controller **150** can detect a “real-time” battery status (or a “charge state”) of each battery pack in a set of battery packs of a battery coupled to the trailer **140**. The controller **150** can then detect a charge state of the battery **106** (e.g., a numerical value, a percentage, a level) as a condition of the trailer **140** and leverage this charge state of the battery **106** to selectively adjust the target preload force.

(183) For example, during a first time period, the controller **150** can: detect a first longitudinal force applied to the kingpin **110** by the hitch of the tow vehicle; detect a first lateral force applied to the kingpin **110** by the hitch; detect a first direction of motion and a first speed of the trailer **140** (e.g., a forward direction); detect a first charge state of a battery pack (e.g., 55%); detect a first tractor-trailer **140** angle; calculate a first total force, applied to the kingpin **110** by the hitch, based on the first longitudinal force and the first lateral force; calculate a first target preload force opposite the first direction of motion (e.g., a reverse direction) proportional the tractor-trailer **140** angle and inversely proportional the first charge state of the battery pack; and, in response to the first total force falling below the first target preload force, trigger the motor **131** to reduce torque output in the first direction of motion (e.g., forward direction) to decrease a first difference between the first total force and the first target preload force.

(184) During a second time period, the controller **150** can: detect a second longitudinal force applied to the kingpin **110** by the hitch of the tow vehicle; detect a second lateral force applied to the kingpin **110** by the hitch; detect a second direction of motion and a second speed of the trailer **140** (e.g., a forward direction); detect a second tractor-trailer **140** angle less than the first tractor-trailer **140** angle; detect a second charge state of the battery pack (e.g., 75%) greater than the first charge state of the battery pack (e.g., 55%); calculate a second total force, applied to the kingpin **110** by the hitch, based on the second longitudinal force and the second lateral force; calculate a second target preload force opposite the second direction of motion (e.g., a reverse direction) proportional the second tractor-trailer **140** angle and inversely proportional the second charge state of the battery pack (e.g., 75%); and, in response to the second total force exceeding the target preload force, trigger the motor **131** to reduce torque output in the second direction of motion (e.g., forward direction) to decrease a second difference between the first total force and the first target

preload force.

(185) Alternatively, in response to the charge state of the battery pack falling below a threshold charge state (e.g., a predefined charge state), the controller **150** can: enter a drag charging mode; calculate a target preload force opposite the direction of motion (e.g., a reverse direction) proportional the tractor-trailer angle and inversely proportional the charge state of the battery pack (e.g., increase target preload force); and, in response to the total force falling below the target preload force, trigger the motor **131** to reduce torque output in the direction of motion (e.g., forward direction) to decrease a difference between the first total force and the first target preload force.

(186) Additionally or alternatively, the controller **150** can access the drive route assigned to the trailer **140** and predict a charge state of the battery **106** at the start of the tow mode. Then for each leg of the drive route, the controller **150** can autonomously increase or decrease the target preload force at the kingpin **110** proportional the predicted charge state of the battery **106** associated with each leg of the drive route.

(187) Therefore, the controller **150** can leverage a “real-time” charge state or a predicted charge state of each battery pack to selectively increase or decrease the target preload force and thereby, increase the life of the battery **106** and reduce emissions by the tow vehicle.

(188) 10.5 Target Preload Force: Weight on Driven Axle

(189) In one variation, the drive system **130** can further include an air-ride suspension system coupled to the driven axle **137** and a pressure sensor configured to output signals representing combined air pressures of air bags in the air-ride suspension system. The controller **150** can then leverage these combined air pressures to approximate a weight of the trailer **140** (e.g., a load, a payload) on the driven axle **137** and then selectively trigger the motor **131** to disable torque output to the driven axle **137** and/or to regeneratively brake the driven axle **137** to slow motion of the trailer **140**.

(190) For example, the drive system **130** can include an air-ride suspension system coupled to the driven axle **137** and includes: a first air bag arranged proximal the left driven wheel **138**; and a second air bag arranged proximal the right driven wheel **139**. The drive system **130** can further include a pressure sensor coupled to the driven axle **137** and configured to output a signal representing a combined air pressure in the first air bag and the second air bag. The controller **150** can then: detect a force applied to the kingpin **110** by a hitch of a tow vehicle; detect a direction of motion of the trailer **140** (e.g., forward direction); and access a target preload force. The controller **150** can detect a combined air pressure in the first air bag and the second air bag based on the signal output by the pressure sensor **115**. Then, in response to the combined air pressure in the first air bag and the second air bag exceeding a threshold air pressure and in response to the force falling below the target preload force, the controller **150** can: trigger the motor **131** to disable torque output to the driven axle **137**; and trigger the motor **131** to regeneratively brake the driven axle **137**.

(191) In another variation, the controller **150** can: detect an air pressure of an air-ride suspension system coupled to the driven axle **137** (e.g., via a pressure sensor); and approximate a weight (e.g., a load, a payload) of the trailer **140** on the driven axle **137** based on the air pressure. Further, a user (e.g., an operator, a manager, a driver) can define a target weight (e.g., a weight limit, a weight standard) for each driven axle **137** and the controller **150** can then leverage this target weight to adjust target regenerative braking proportional to the approximated weight of the trailer **140** on the driven axle **137**.

(192) For example, the controller **150** can: detect a force applied to the kingpin **110** by a hitch of a tow vehicle; detect a direction of motion of the trailer **140** (e.g., forward direction); and access a target preload force. The controller **150** can then: interpret a weight of the trailer **140** on the driven axle **137** (e.g., 34,500 pounds) based on the combined air pressure; and, in response to the weight (e.g., 34,500 pounds) exceeding a target weight for the driven axle **137** (e.g., 34,000 pounds) and in

response to the force exceeding the target preload force, trigger the motor **131** to regeneratively brake the driven axle **137** to redistribute the weight throughout the trailer **140**.

(193) Therefore, the controller **150** can interpret air pressures from the air-ride suspension system of the trailer **140** and leverage these air pressures to trigger the motor **131** to disable torque output to the driven axle **137** and/or to regeneratively brake the driven axle **137** to slow motion of the trailer **140**. Additionally, the controller **150** can leverage these air pressures to interpret a weight of the trailer **140** on the driven axle **137** and trigger the motor **131** to regeneratively brake the driven axle **137** in order to achieve a weight balance of the trailer **140** on the driven axle **137**.

(194) 10.6 Regenerative Braking: Wheel Speed Sensors+Traction

(195) In one variation, the system **100** can include a set of wheel speed sensors. Each wheel speed sensor can couple to a corresponding driven wheel of the drive system **130** and the controller **150** can leverage signals output from each wheel speed sensor to track traction between the driven wheels of the drive system **130** and a ground surface and detect a speed of the trailer **140**. The controller **150** can then trigger the motor **131** to selectively increase or decrease regenerative braking of the left driven wheel **138** and the right driven wheel **139** based on the level of traction and/or the speed of the trailer **140**.

(196) For example, the controller **150** can: detect a force applied to the kingpin by the hitch; and access a target preload force on the kingpin **110**. Then, in response to the force exceeding the target preload force, the controller **150** can trigger the motor **131** to regeneratively brake the trailer **140**. The controller **150** can then: detect loss of traction at the driven axle **137** (e.g., between the left driven wheel **138** and a ground surface) based on a signal output by the set of wheel speed sensors; and, in response to detecting loss of traction at the driven axle **137** and in response to the force exceeding the target preload force, trigger the motor **131** to reduce regenerative braking to the driven axle **137**.

(197) Further, the controller **150** can: detect a presence of traction at the driven axle **137** (e.g., between the left driven wheel **138** and the ground surface) based on a next signal output by the set of wheel speed sensors; and, in response to detecting presence of traction at the driven axle **137**, automatically trigger the motor **131** to increase regenerative braking to the driven axle **137** (e.g., the left driven wheel **138**).

(198) Therefore, the controller **150** can leverage the target preload force and signals output from the set of wheel speed sensors to regeneratively brake the driven axle **137**, and thereby, maintain traction between the driven wheels of the trailer **140** and a ground surface.

(199) 10.7 Unauthorized Movements of Trailer

(200) In one variation, the controller **150** can detect a force applied to the kingpin **110** in a direction opposite an acceleration of the trailer **140** and identify an unauthorized movement of the trailer **140**. The controller **150** can then trigger the motor to disable torque output and trigger the drive system **130** to apply a regenerative brake force to the trailer **140**.

(201) For example, the controller **150** can: access a second signal from a set of force sensors **119** (e.g., strain gauges); calculate a direction of the second force based on the signal; in response to detecting the direction of the second force opposite to the first direction of a first force (e.g., direction of force opposite acceleration of the trailer **140**), trigger the motor to disable torque output; and trigger the drive system **130** to apply a regenerative brake force proportional to the second force.

(202) Additionally, in response to detecting the direction of the second force opposite to the first direction of a first force (e.g., direction of force opposite acceleration of the trailer **140**), the controller **150** can: identify the direction of the second force opposite to the first direction of the first force as an unauthorized movement event of the trailer **140**; and generate an alert notifying a user (e.g., an operator, an owner, a driver) of this unauthorized movement event.

(203) Therefore, in tow mode, the controller **150**, the kingpin **110**, and the drive system **130** can cooperate to prevent unauthorized movement events of the trailer **140** by: monitoring forces and

accelerations of the trailer **140** at the kingpin **110**; disabling torque output by the motor; and applying a regenerative brake force to slow motion of the trailer **140**.

(204) 11. Variation: Sensor Redundancy+Fasteners with Integral Sensors

(205) In one variation, the kingpin **110** can include a set of sensor fasteners configured to mount the kingpin **110** to the trailer **140** and to output signals corresponding to lateral and longitudinal forces applied to the kingpin **110**.

(206) In one implementation, the set of fasteners are configured to fasten the base of the kingpin **110** to a floor **146** of the trailer **140** via the set of through-bores **114** and include: a first sensor fastener including a first strain gauge and arranged in a first through-bore, in the set of through-bores **114**, laterally offset from the kingpin **110**; and a second sensor fastener including a second strain gauge and arranged in a second through-bore, in the set of through-bores **114**, longitudinally offset from the kingpin **110**.

(207) In another implementation, the set of fasteners is configured to fasten the base of the kingpin **110** to a floor **146** of the trailer **140** via the set of through-bores **114** and includes a first sensor fastener and a second sensor fastener. The first sensor fastener includes a first sensor arranged in a first through-bore, in the set of through-bores **114**, laterally offset from the kingpin **110** and configured to output a signal representing a first vertical load through the kingpin **110** proximal the first through-bore. The second sensor fastener includes a second sensor arranged in a second through-bore, in the set of through-bores **114**, longitudinally offset from the kingpin **110** and configured to output a signal representing a second vertical load through the kingpin **110** proximal the second through-bore.

(208) Furthermore, the controller **150** can access signals from the set of strain gauges and implement Blocks of the method **S100** to calculate a direction and a magnitude of a force applied to the kingpin **110**. Similarly, the controller **150** can access signals from the set of sensor fasteners and implement Blocks of the method **S100** to calculate the direction and the magnitude of this force applied to the kingpin **110**. The controller **150** can then leverage the direction and magnitude of the force from the set of sensors **119** and from the set of sensor fasteners to increase torque output and/or to identify a failure of a strain gauge at the kingpin **110**.

(209) For example, the controller **150** can: access a first signal from the first sensor **119** representing a first lateral force applied to the kingpin **110**; access a second signal from the second sensor **119** (e.g., a strain gauge) representing a first longitudinal force applied to the kingpin **110**; and calculate a first direction and a first magnitude of a first force applied to the kingpin **110** based on the first signal and the second signal. The kingpin **110** can include the set of fasteners **118** configured to fasten the base of the kingpin **110** to the floor **146** of the trailer **140** via the set of through-bores **114**. The set of fasteners **118** can include: a first sensor fastener arranged in a first through-bore, in the set of through-bores **114**, laterally offset from the kingpin **110** and including a third sensor configured to output a third signal representing a first vertical load through the kingpin **110** proximal the first through-bore; and a fourth sensor fastener arranged in a second through-bore, in the set of through-bores **114**, longitudinally offset from the kingpin **110** and including a second sensor configured to output a fourth signal representing a second vertical load through the kingpin **110** proximal the second through-bore. The controller **150** can then: access the third signal from the first sensor fastener; interpret a second lateral force applied to the kingpin **110** based on the third signal; access the fourth signal from the second sensor fastener; and interpret a second longitudinal force applied to the kingpin **110** based on the fourth signal. At a first time in tow mode, the controller **150** can trigger the motor **131** to output the torque in the first direction of the first force and proportional the first magnitude of the first force, in response to correspondence between the first lateral force and the second lateral force, and in response to correspondence between the first longitudinal force and the second longitudinal force. At a second time, in tow mode, the controller **150** can deactivate the motor in response to incongruity between the first longitudinal force and the second longitudinal force.

(210) In another variation, the kingpin **110** includes a set of sensors **119** configured to generate analog voltages of forces applied to the kingpin **110** and an amplifier configured to convert these analog voltages into digital signals. The controller **150** can then leverage these digital signals to calculate a direction and a magnitude of a force applied to the kingpin **110** and compare this direction and magnitude to analog values from the force sensors **119**.

(211) For example, the kingpin **110** can include: a first sensor **119** configured to output a first signal as a first analog voltage corresponding to a lateral force applied to the kingpin **110**; and a second sensor **119** configured to output a second signal as a second analog voltage corresponding to a longitudinal force applied to the kingpin **110**. The controller **150** can then calculate a first direction of a first force applied to the kingpin **110** based on the first analog voltage and the second analog voltage. The kingpin **110** can also include an amplifier configured to: convert the first analog voltage into a first digital signal within a voltage range representing the lateral force; and convert the second analog voltage into a second digital signal within the voltage range representing the longitudinal force. The controller **150** can: calculate a second direction of the first force applied to the kingpin **110** based on the first digital signal and the second digital signal; detect the first direction deviating from the second direction of the first force; in response to detecting the first direction deviating from the second direction of the first force for a duration of time exceeding a threshold duration of time, trigger the motor **131** to disable torque output in Block **S180**; and activate a hibernation mode. Thus, the controller **150** can verify the direction and magnitude of a force applied to the kingpin **110** with analog values and/or digital signals to identify sensor **119** failures at the kingpin **110** and enter a hibernation mode.

(212) Therefore, the controller **150** can leverage the set of strain gauges of the kingpin **110** and the set of sensor fasteners to identify a strain gauge failure at the kingpin **110** and to selectively deactivate the motor **131**.

(213) However, the kingpin **110** can include any other type of sensor fastener in order to fasten the kingpin **110** to the floor **146** of a trailer **140** and output signals representing forces applied to the kingpin **110**.

(214) 11.1 Coupling Event Confirmation+Proximity Sensors

(215) In one variation, shown in FIGS. **6D** and **6E**, the kingpin **110** can include a set of proximity sensors (e.g., Hall effect sensors, capacitive sensors, inductive sensors) configured to output signals representing presence of a hitch of a tow vehicle within a threshold distance of the kingpin **110**. The controller **150** can then leverage these signals to interpret presence of the hitch within the threshold distance of the kingpin **110** and confirm a coupling event with the kingpin **110**.

Alternatively, the controller **150** can leverage these signals to detect absence of the hitch within the threshold distance of the kingpin **110** and, in response to detecting absence of the hitch within the threshold distance of the kingpin **110**, trigger the motor **131** to disable torque output to prevent a jackknifing event.

(216) For example, the kingpin **110** can include an array of proximity sensors: coupled to and radially arranged on the base of the kingpin **110**; facing the shank of the kingpin **110**; and configured to output a signal representing absence of a hitch of the tow vehicle (e.g., jaws of a fifth wheel of a tractor) within a threshold distance of the kingpin **110**. The controller **150** can then: access the signal; detect absence of the hitch of the tow vehicle (e.g., jaws of a fifth wheel of a tractor) within the threshold distance of the kingpin **110**; and, in response to detecting absence of the hitch within the threshold distance of the kingpin **110**, trigger the motor **131** to disable torque output, and thus, prevent a jackknifing event between the tow vehicle and the trailer **140** (e.g., tow vehicle brakes to a stop and the trailer **140** continues to move toward and push the tow vehicle kingpin **110**).

(217) Alternatively, the controller **150** can: access the signal; detect presence of the hitch of the tow vehicle (e.g., jaws of a fifth wheel of a tractor) within the threshold distance of the kingpin **110**; and, in response to detecting presence of the hitch within the threshold distance of the kingpin **110**,

confirm a coupling event between the hitch of the tow vehicle and the trailer **140**.

(218) The systems and methods described herein can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions can be executed by computer-executable components integrated with the application, applet, host, server, network, website, communication service, communication interface, hardware/firmware/software elements of a user computer or mobile device, wristband, smartphone, or any suitable combination thereof. Other systems and methods of the embodiment can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions can be executed by computer-executable components integrated by computer-executable components integrated with apparatuses and networks of the type described above. The computer-readable medium can be stored on any suitable computer readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, or any suitable device. The computer-executable component can be a processor but any suitable dedicated hardware device can (alternatively or additionally) execute the instructions.

(219) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

Claims

1. A method comprising: during a first time period: detecting a first load of a trailer; detecting a first acceleration of the trailer; deriving a first representation of force applied to a kingpin, arranged on a proximal end of the trailer, by a hitch of a tow vehicle based on the first acceleration and the first load of the trailer; accessing an incline angle and a first direction of motion of the trailer; defining a first target preload force opposite the first direction of motion and proportional to the first incline angle; and in response to the first representation of force falling below the first target preload force, triggering a motor, arranged in a drive system located proximal a distal end of the trailer, to increase torque output in the first direction of motion to decrease a first difference between the first representation of force and the first target preload force; and during a second time period: detecting a second acceleration of the trailer; deriving a second representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the second acceleration and the first load of the trailer; accessing a decline angle of the trailer; defining a second target preload force opposite the first direction of motion and inversely proportional to the decline angle; and in response to the second representation of force falling below the second target preload force, triggering the motor to increase torque output opposite the first direction of motion to decrease a second difference between the second representation of force and the second target preload force.
2. The method of claim 1: wherein detecting the first load of the trailer comprises: accessing a first signal from a pressure sensor coupled to the drive system supporting a distal end of the trailer; detecting an air pressure within a pneumatic suspension system supporting the drive system based on the first signal; and interpreting the first load of the trailer carried by the drive system based on the air pressure within the pneumatic suspension system; and wherein detecting the first acceleration of the trailer comprises: accessing a second signal representing accelerations of the trailer from an accelerometer arranged on the trailer proximal the kingpin; and detecting the first acceleration of the trailer based on the second signal.
3. The method of claim 1: wherein accessing the first direction of motion of the tow vehicle comprises interpreting the first direction of motion of the trailer in a forward direction; wherein defining the first target preload force comprises calculating the first target preload force, in a reverse direction, inversely proportional to the incline angle, the reverse direction opposite the forward direction; wherein triggering the motor to reduce torque output in the first direction of

motion during the first time period comprises triggering the motor to increase torque output in the forward direction to decrease the first difference between the first representation of force applied to the kingpin and the first target preload force; and wherein triggering the motor to decrease torque output in the first direction of motion during the second time period comprises triggering the motor to decrease torque output in the forward direction to decrease the second difference between the second representation of force applied to the kingpin and the second target preload force.

4. The method of claim 1: further comprising, during the first time period: interpreting a first speed of the trailer; accessing a first signal representing a first distance between the trailer and a left feature on a corresponding left side of the tow vehicle; accessing a second signal representing a second distance between the trailer and a right feature on a corresponding right side of the tow vehicle; and calculating a first tractor-trailer angle based on the first signal and the second signal; wherein accessing the first target preload force comprises calculating the first target preload force: opposite the first direction of motion; proportional to the first tractor-trailer angle; and inversely proportional to the first incline angle; and wherein triggering the motor to increase torque output in the first direction of motion during the first time period comprises, in response to the first speed of the trailer falling below a first threshold speed and in response to the first representation of force applied to the kingpin falling below the first target preload force: triggering the motor to reduce torque output in the first direction of motion to decrease the first difference between the first representation of force applied to the kingpin and the first target preload force.

5. The method of claim 4: further comprising, during the second time period: accessing a third signal representing a third distance between the trailer and the left feature on the corresponding left side of the tow vehicle; accessing a fourth signal representing a fourth distance between the trailer and the right feature on the corresponding right side of the tow vehicle; calculating a second tractor-trailer angle based on the third signal and the fourth signal, the second tractor-trailer angle less than the first tractor-trailer angle; and interpreting a second speed of the trailer; wherein defining the second target preload force during the second time period comprises calculating the second target preload force: approximating the first target preload force; proportional to the second tractor-trailer angle; and inversely proportional to the decline angle; and wherein triggering the motor to increase torque output opposite the first direction of motion to decrease the second difference during the second time period comprises in response to the second speed exceeding the first threshold speed and in response to the second representation of force falling below the second target preload force: triggering the motor to increase torque output opposite the first direction of motion to decrease the second difference between the second representation of force and the second target preload force.

6. The method of claim 1: further comprising, during the first time period: accessing a first signal representing a first wheel speed of the trailer from a first wheel speed sensor; accessing a second signal representing a second wheel speed of the trailer from a second wheel speed sensor; and calculating a first wheel speed difference of the trailer based on the first signal and the second signal; wherein defining the first target preload force comprises defining the first target preload force: opposite the first direction of motion; and proportional to the first wheel speed difference; further comprising, during the second time period: accessing a third signal representing a third wheel speed of the trailer from the first wheel speed sensor; accessing a fourth signal representing a fourth wheel speed of the trailer from the second wheel speed sensor; and calculating a second wheel speed difference of the trailer based on the third signal and the fourth signal, the second speed difference greater than the first speed difference; and wherein defining the second target preload force comprises defining the second target preload force: opposite the first direction of motion; and proportional to the second speed difference.

7. The method of claim 1: further comprising, during the first time period, detecting a first longitudinal force applied to the kingpin by the hitch; wherein deriving the first representation of force applied to the kingpin comprises calculating a first total force applied to the kingpin based on

the first longitudinal force, the first acceleration, and the first load; further comprising, during the second time period, detecting a second longitudinal force applied to the kingpin by the hitch; and wherein deriving the second representation of force applied to the kingpin comprises calculating a second total force applied to the kingpin based on the second longitudinal force, the second acceleration, and the first load.

8. The method of claim 1, further comprising, during a third time period: detecting a third acceleration of the trailer; deriving a third representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the third acceleration and the first load of the trailer; accessing a second decline angle of the trailer; defining a third target preload force: opposite the first direction of motion; and proportional to the second decline angle; and in response to the third representation of force exceeding the third target preload force, triggering the motor to regeneratively brake a driven axle of the drive system: to slow motion of the trailer; and to decrease a third difference between the third representation of force and the third target preload force.

9. The method of claim 8: wherein accessing the first direction of motion of the tow vehicle comprises interpreting the first direction of motion of the trailer in a forward direction; wherein defining the third target preload force opposite the first direction of motion comprises calculating the third target preload force in the reverse direction, opposite the forward direction, and proportional to the second decline angle of the trailer; and wherein triggering the motor to regeneratively brake the driven axle to slow motion of the trailer comprises triggering the motor to regeneratively brake the driven axle of the drive system: to increase torque output in the reverse direction; and to decrease the third difference between the third representation of force and the third target preload force.

10. The method of claim 1: wherein detecting the first load of the trailer comprises: accessing a first signal output by a load cell coupled to the kingpin; interpreting the first load of the trailer based on the first signal; and wherein detecting the first acceleration of the trailer comprises: accessing a second signal output by an accelerometer arranged on the trailer proximal the kingpin; and detecting the first acceleration of the trailer based on the second signal.

11. The method of claim 1, wherein triggering the motor to increase torque output opposite the first direction of motion during the second time comprises, in response to the second representation of force falling below the second target preload force, triggering the motor to regeneratively brake a driven axle of the drive system: to increase torque output opposite the first direction of motion; and to decrease the second target preload force.

12. The method of claim 1: further comprising, during the first time period, detecting a first charge state of a battery coupled to the trailer; wherein defining the first target preload force comprises calculating the first target preload force: opposite the first direction of motion; proportional to the incline angle; and inversely proportional to the first charge state; further comprising, during the second time period, detecting a second charge state, greater than the first charge state, of the battery; and wherein defining the second target preload force comprises calculating the second target preload force: approximating the first target preload force; inversely proportional to the decline angle; and inversely proportional to the second charge state.

13. The method of claim 1, further comprising, during a third time period: detecting a third acceleration of the trailer; detecting a second load of the trailer, the second load less than the first load of the trailer; deriving a third representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the third acceleration and the second load of the trailer; interpreting a second incline angle of the trailer; defining a third target preload force opposite the first direction of motion and proportional to the second incline angle; and in response to the third target representation of force falling below the third target preload force, triggering the motor to increase torque output in the first direction of motion to decrease a third difference between the third representation of force and the third target preload force.

14. The method of claim 1: wherein detecting the first acceleration of the trailer comprises

detecting a first change in acceleration of the trailer; wherein defining the first target preload force opposite the first direction of motion comprises calculating a first change in torque output in the first direction of motion based on the first change in acceleration; wherein triggering the motor to increase torque output in the first direction of motion during the first time period comprises triggering the motor to increase torque output to the first change in torque output; wherein detecting the second acceleration of the trailer comprises detecting a second change in acceleration of the trailer; wherein defining the second target preload force opposite the first direction of motion comprises calculating a second change in torque output opposite the first direction of motion based on the second change in acceleration; and wherein triggering the motor to increase torque output opposite the first direction of motion during the second time period comprises triggering the motor to increase torque output to the second change in torque output.

15. A method comprising: during a first time period: detecting a first acceleration of a trailer; deriving a first representation of force applied to a kingpin, arranged on a proximal end of the trailer, by a hitch of a tow vehicle, based on the first acceleration of the trailer; accessing a direction of motion of the trailer; detecting a first wheel speed difference of the trailer; defining a first target preload force opposite the direction of motion and proportional to the first wheel speed difference; and in response to the first representation of force falling below the first target preload force, triggering a motor, arranged in a drive system located proximal a distal end of the trailer, to increase torque output in the direction of motion to decrease a first difference between the first representation of force and the first target preload force; and during a second time period: detecting a second acceleration of the trailer; deriving a second representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the second acceleration of the trailer; detecting a second wheel speed difference of the trailer; defining a second target preload force opposite the direction of motion and proportional to the second wheel speed difference; and in response to the second representation of force falling below the second target preload force, triggering the motor to increase torque output opposite the direction of motion to decrease a second difference between the second representation of force and the second target preload force.

16. The method of claim 15: further comprising, during the first time period, detecting a first duration of the trailer exhibiting the first wheel speed difference; wherein defining the first target preload force comprises defining the first target preload force opposite the direction of motion of the trailer, proportional to the first wheel speed difference, and inversely proportional to the first duration; and further comprising, during a third time period: detecting a third wheel speed difference, corresponding to the first wheel speed difference, of the trailer; detecting a second duration of the trailer exhibiting the first wheel speed difference; detecting a third acceleration of the trailer; deriving a third representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the third acceleration; in response to the second duration exceeding the first duration, defining a third target preload force: less than the first target preload force; opposite the direction of motion; and inversely proportional to the second duration; and in response to the third representation of force falling below the third target preload force, triggering the motor to reduce torque output in the direction of motion to decrease a third difference between the third representation of force and the third target preload force.

17. The method of claim 16: further comprising, during the first time period, detecting a first load of the trailer; wherein deriving the first representation of force applied to the kingpin during the first time period comprises deriving the first representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the first acceleration of the trailer and the first load of the trailer; wherein deriving the second representation of force applied to the kingpin during the second time period comprises deriving the second representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the second acceleration of the trailer and the first load of the trailer; and wherein deriving the third representation of force applied to the kingpin during the third time period comprises, deriving the third representation of force applied to the kingpin by the hitch of

the tow vehicle, based on the third acceleration and the first load of the trailer.

18. The method of claim 15: wherein detecting the first wheel speed difference of the trailer comprises: accessing a first signal representing a first wheel speed of the trailer from a first wheel speed sensor; accessing a second signal representing a second wheel speed of the trailer from a second wheel speed sensor; and detecting the first wheel speed difference of the trailer based on the first signal and the second signal; and wherein detecting the second wheel speed difference of the trailer comprises: accessing a third signal representing a third wheel speed of the trailer from the first wheel speed sensor; accessing a fourth signal representing a fourth wheel speed of the trailer from the second wheel speed sensor; and detecting the second wheel speed difference of the trailer based on the third signal and the fourth signal, the second wheel speed difference greater than the first wheel speed difference.

19. A method comprising: during a first time period: detecting a first direction and a first change in acceleration of the trailer; accessing an incline angle and a direction of motion of the trailer; calculating a first change in torque output in the direction of motion based on the first change in acceleration; and triggering a motor, arranged in a drive system located proximal a distal end of the trailer, to increase torque output in the direction of motion to the first change in torque output; and during a second time period: detecting the direction of motion and a second change in acceleration of the trailer; calculating a second change in torque output opposite the direction of motion based on the second change in acceleration; and triggering the motor to increase torque output opposite the direction of motion to the second change in torque output.

20. The method of claim 19: further comprising, during the first time period: detecting a first load of the trailer; deriving a first representation of force applied to a kingpin, arranged on a proximal end of the trailer, by a hitch of a tow vehicle based on the first change in acceleration and the first load of the trailer; and wherein calculating the first change in torque output in the direction of motion comprises defining a first target preload force: opposite the direction of motion; and proportional to the incline angle; and wherein triggering the motor to increase torque output in the direction of motion during the first time period comprises, in response to the first representation of force falling below the first target preload force: triggering the motor to increase torque output in the direction of motion to decrease a first difference between the first representation of force and the first target preload force; and further comprising, during the second time period: deriving a second representation of force applied to the kingpin, by the hitch of the tow vehicle, based on the second change in acceleration and the first load of the trailer; and accessing a decline angle of the trailer; wherein calculating the second change in torque output in the direction of motion comprises defining a second target preload force: opposite the direction of motion; and inversely proportional to the decline angle; and wherein triggering the motor to increase torque output opposite the direction of motion during the second time period comprises, in response to the second representation of force falling below the second target preload force: triggering the motor to increase torque output opposite the direction of motion to decrease a second difference between the second representation of force and the second target preload force.
